

Hamzstlab Mathematics and Hamzstplot

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¹A thank you or further information

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Preface

For my Wife Freya, and our daughters Solreya, Mithra, Catenary, Iyzumrae and Zefir.

For Lucrif and Znane too along with all the 8 Queens.

For Albert Silverberg, Camus and Miklotov who left TREVOCALL to guard me.

To Nature(Kala, Kathmandu, Big Tree, Sentinel, Aokigahara, Hoia Baci, Jacob's Well, Mt Logan, etc) and my family Berlin: I have served, I will be of service.

To my dogs who always accompany me working in Valhalla Projection, go to Puncak Bintang or Kathmandu: Sine Bam Bam, Kecil, Browni Bruncit, Sweden Sexy, Cambridge Klutukk, Milan keng-keng, Piano Blutut and more will be adopted. To my cat who guard the home while I'm away with my dogs: London.

The one who moves a mountain begins by carrying away small stones - Confucius

A book to explained how we create a C++ library from forking / branching out ImGUI 1.92.0 WIP, Implot, Implot 3D, merge them into one big pile that we call Hamzstlab Mathematics so it can be easily compiled / build so we can focus more on the Mathematics side / learning.

We also write about Hamzstplot that is forking / branching out from Matplot++, it is a C++ plotting library with backend gnuplot, so you can have more creativity in plotting since the syntax will be all C++. It is not as interactive as Hamzstlab Mathematics but it has more options of plotting that you can play with.

a little about GlanzFreya:

Freya the Goddess is (DS Glanzsche') Goddess wife. We get married on Puncak Bintang on November 5th, 2020 after we go back from Waghete, Papua. My wife birthday (Freya the Goddess) is on August 1st. After not working with human anymore since 2019, I (DS Glanzsche) work in a forest, shaping a forest into a natural wonder / like a canvas for painting, and also clean up trashes that are thrown in the forest. Thus after 6 years working with Nature I have found what can makes me waltz to work, it is going to the forest in the morning then go back home again and study / learn science, engineering, computer programming.

Freya specialty is Electrical Engineering, she is the one that taught Glanz about Arduino Uno and all that.

Glanz specialty is Dogs, she loves dogs and love feeding stray dogs and taking home stray puppies given she has money and foods, otherwise she will bring stray dogs to animal shelter.

a little about Hamzst:

Hamzst is a nickname for Alice from Persona game series made by Atlus, sometimes she can also be similar with Alice from Alice in Wonderland (1951 movie). When she said "But in my world, the books will be nothing but picture," it is true for Hamzstplot and Hamzstlab Mathematics. She is a Ding, Yin Fire daymaster (The candle), she is the controller of Glanz and bring a lot of luck to Glanz too. Her specialty is Physics.

a little about Lasthrim:

She was known as Anabelle the Doll from the famous movie "The Conjuring," but she changes her name into Lasthrim since 2021 and want a true repentance since she has found a new joy and Love in mathematics. She is a Bing, Yang Fire daymaster (The Sun), she can shapes Glanz into the best Excalibur. Her specialty is Mathematics. Realizing that after writing Lasthrim Projection book, it is better to use our own C++ library modifying from existing C++ library to make the plot, computation and simulation, instead of using Python and JULIA. So here she is.



Figure 1: *Freya, thank you for everything, I am glad I marry you and I could never have done it without you.*



Figure 2: *I paint her 3 days before Christmas in 2021.*

Chapter 1

Introduction Story

On top of the mountain a heart shaped stone waiting for You, my eyes can't see, my body can't touch, but my heart knows it's You. Forever only You. - Glanz to Freya

This book is written on May 29th, 2025, while we also have another project to make SymIntegration C++ library, we are working on this when we saw Implot, Implot 3D, and Matplot++. We want to clone them and modify them, or probably add some features to plot, simulate and more in the future, it is called branching out / forking. Then rename it as Hamzstlab Mathematics for combining Imgui+Implot+Implot3D+Imnodes, as a C++ library / software where you can create interactive graph or simulation for mathematics problems. It will be very useful for students and researchers. While Hamzstplot is branching out from Matplot++ to create graph and plot as well but it is not interactive as Hamzstlab Mathematics but the options for building the plot are enormous. We can choose which one to use depends on our need.

To avoid confusion in the future: Whenever there is a written of subject "I" it means DS Glanzsche as the I subject.

Years ago, around December 2019 when I (DS Glanzsche) was hiking / jogging up to Puncak Bintang, I pray at dawn, I remember I used to arrive there at 4:30 AM , departing around 3:00 AM from home, what I pray is " I pray and hope one day I can create a software like MATLAB," well I am starting this and remember that prayer again, it is becoming true, when I was installing MATLAB back then in 2011 to 2014 for college purpose, they give MATLAB course to students and even pay the license for it, I think it is great, but then realize, when I was reading a book like "Elementary Linear Algebra" or ""Elementary Differential Equation" or "Calculus" none of them have the MATLAB code to show how the plot is made or the computation is being computed, even if today there is a book called "Advanced Engineering Mathematics with MATLAB" we still need the code for the basic that undergraduate needs, since the knowledge for other science and engineering department will rely heavily on those Calculus, Linear Algebra, and Differential Equations first.

Not to mention, that there are tons of menu / on the menu bar on MATLAB, that the user won't even need to use for the rest of his/her life, the size of MATLAB today is over 20 GB, and for the installation it takes hours, now I am comparing with Hamzstlab Mathematics, if you are an Electrical Engineer, and we already create the basic code and API to create any electric circuit with basic NPN' or PNP' BJT transistor or MOSFET n-channel or p-channel along with any number of resistors, inductor, and capacitor and will have the computed voltage and current, and you

probably won't even need to download Hamzstlab Mathematics for even 1 GB, isn't it going to save your day and your time?

This is the real life example where we try to plot a solution to a simple first order ordinary differential equation and the Euler' method approximation, the size of the C++ file and the Makefile only consumes 3.1 KB, and if you build the example the C++ codes it will create this object files that will consumes 27.3 MB of size of your storage, which can be cleaned after you are finish. Hamzstlab Mathematics will only consume your storage of 50.3 MB, and the rest will depends on the project that you will be working on.



Figure 1.1: The interactive plot of Euler' method approximation and the solution plot for $\frac{dy}{dt} = 3 - 2t - \frac{1}{2}y$ with Hamzstlab Mathematics.

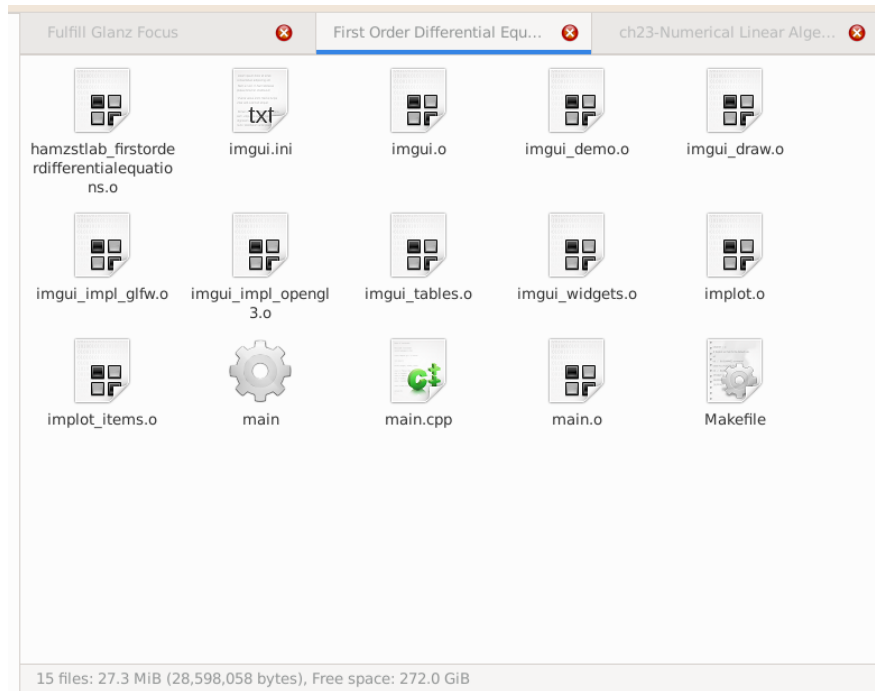


Figure 1.2: The size of a built example with all the object files.

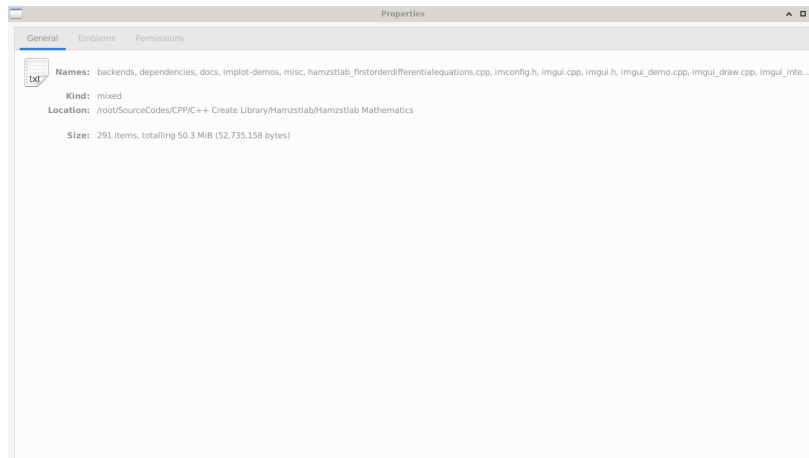


Figure 1.3: The size of Hamzslab Mathematics source code as of June 4th, 2025.

On May 26th, 2025 while in St Peter Cathedral in Bandung I was reading a book [5], then at the first chapter it stated like this:

If all you are interested in is a quick calculation, then Python along with its ecosystem is likely going to be your one-stop shop. As your work becomes more computationally challenging, you may need to switch to a compiled language; most work on supercomputers is carried out using languages like Fortran or C++ or sometimes even C.

Basically, what ImGui combined with Implot and Implot 3D now able to do, show a lot of

interactive visualizations in one window, is a big gap between C++ and Python, like the sky and the earth. I (DS Glanzsche) as one of the author, starts learning C++ rigorously since October 2023, still very infant stage, then able to write DianFreya Math Physics Simulator and already finish the whole Elementary Linear Algebra book along with all the C++ codes for plotting and simulation. Previously, I was using Python and JULIA to do the computation while writing the book Lasttrim Projection (focusing on Calculus and basic Mathematics field that are more theoretical like Discrete Mathematics, Elementary Differential Equations, Partial Differential Equations). Now I have learned Python, JULIA, C++, Fortran, and C. I know writing a code with C++ to achieve the same result will take longer and steeper learning curve, but it is worth the time spent.

The Operating System and Softwares in Use:

1. GFreya OS version 1.8 (based on LFS and BLFS version 11.0 System V books)
2. GCC version 11.2.0 (to compile C++ codes)
3. Hamzstplot version 1.5
4. Hamzstlab Mathematics version 1.51

We will explain about Hamzstplot and Hamzstlab Mathematics in the next two chapters, then the rest of the chapters will be talking about their application (in creating the plot, interactive plot, or simulation with either Hamzstplot or Hamzstlab Mathematics).

The content of this book covers the basic undergraduate courses in Mathematics from Differential Equation to Numerical Analysis and Linear Programming. If one is keen and acute enough and wonder, why there is no Elementary Linear Algebra here? Guilty. It is because we have done the summarization of the whole Elementary Linear Algebra based on [9] book in another book [4] that we are working on from 2023 to 2025, and we are mainly using C++ libraries that are already well known like Armadillo, Eigen, and GiNaC to help us for the matrix and vector computation till QR method, spectral decomposition, eigenvalue and eigenvector. We let the reader read that book or any other Elementary Linear Algebra book to get the basic knowledge of Linear Algebra, since it is really needed for almost all fields in mathematics, science and engineering. This books is focusing on showing what Hamzstplot, Hamzstlab Mathematics can do to help compute, plot and simulate the mathematics problems we have in each branch / field. We didn't mention it in the cover but we will also use SymIntegration for the symbolic integration computation part.

We are really appreciate those who spend time to write constructing critics, not hate words without reason that will only add more good luck and karma to me, write their opinions on what should be added in SymIntegration or if there is an incorrect C++ codes or algorithm to compute the integrals. We cannot provide the readers who send nice feedbacks with anything now, but if we have money, when someone send / email me a very good feedbacks then we will send you either cookies, doughnuts or parcel of foods. Feedbacks can be sent to **dsglansche@gmail.com**.

Chapter 2

Hamzstplot

"The best and most beautiful things in the world cannot be seen or even touched - they must be felt with the heart" - Helen Keller

Hamzstplot was born on a Full Moon day on May 12th, 2025. We are branching out / forking from Matplot++ and we want to develop it further, changing the name first and will add several features in the future to help us in learning Mathematics, or other branch of science, e.g. to create direction fields. The basic syntax of C++ will really help for a very long term future prospects and more complex problems in science to be computed, analyzed and plot.

I. BUILD AND INSTALL HAMZSTPLOT IN GFREYA OS

We are using GFreya OS as it is a custom Linux-based Operating System, your distro your rules. The commands here assuming that you / the reader are using Linux-based OS, they are familiar and easy to follow for other Linux-based OS users.

To download Hamzstplot, open terminal / xterm then type:
git clone <https://github.com/glanzkaiser/Hamzstplot.git>
Enter the directory then type:

```
mkdir build
cd build
cmake -DBUILD_SHARED_LIBS=ON ..
make
make install
```

Code 1: *Create Hamzstplot dynamic library*

It will create the dynamic library **libhamzstplot.so**, we are using CMake, and it is said that dynamic library saves more RAM memory than static library.

The prefix at default is **/usr/local** so the installed dynamic library will go to **/usr/local/lib**.

Then open terminal and from the current working directory / this repository main directory:

```
cd /source/3rd_party/cimg/
cp -r CImg.h* /usr/local/include

cd /

cd ../Hamzstplot/source/hamzstplot/
cp -r * /usr/local/include

cd ../Hamzstplot/source/3rd_party/nodesoup/include/
cp -r nodesoup.hpp /usr/local/include

cd /
cd ../Hamzstplot/source/3rd_party/nodesoup/src/
bash dynamiclibrary.sh
cp -r libnodesoup.so /usr/local/lib
```

Code 2: *Move include files and 3rd party library*

When we want to create the shared library it will look for this header files in the default path where the include files are usually located. In Linux OS **usr/include** is the basic / default path, another default path is **usr/local/include**.

All files, examples codes and source codes used in this books can be found in this repository: <https://github.com/glanzkaiser/Hamzstplot>

II. EXAMPLE OF PLOTTING WITH HAMZSTPLOT IN GFREYA OS

We will create a 3D plot of a helix, the source code is this:

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

int main() {
    using namespace hamzstplot;

    auto xt = [](double t) { return sin(t); };
    auto yt = [](double t) { return cos(t); };
    auto zt = [](double t) { return t; };
    fplot3(xt, yt, zt);

    show();
    return 0;
}
```

Code 3: *Hamzstplot/Examples with Makefile/Plot 3D Helix/main.cpp*

Open terminal and from the current working directory **Hamzstplot/Examples with Makefile/-Plot 3D Helix/**, then type:

```
make
./main
```

Code 4: *Build and run the executable*

or you can also type in the current working directory (a manual way to compile):
g++ main.cpp -o main -lhamzstplot
./main

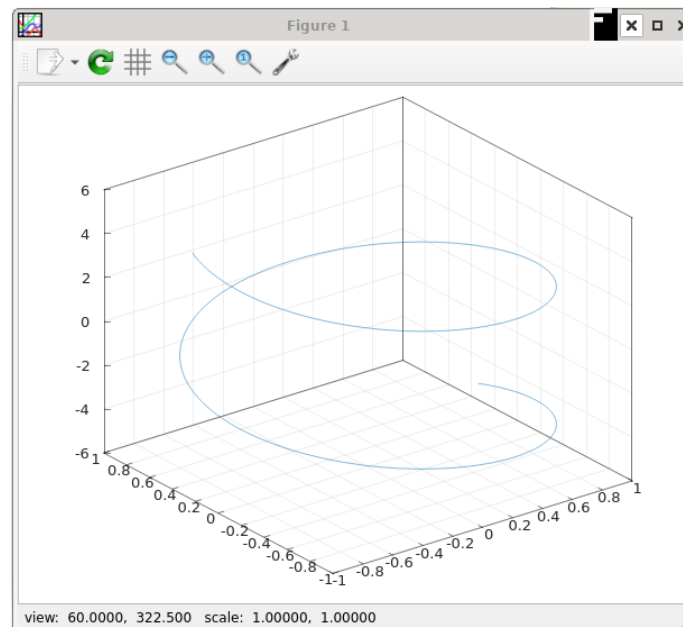


Figure 2.1: The plot here is using the gnuplot as the backend, that is why the GUI window should look very familiar.

Chapter 3

Hamzstlab Mathematics

"In this world perhaps, but in my world, the books will be full of pictures" - Alice (Alice in Wonderland - 1951)

Why would we need Hamzstlab Mathematics if we already have Hamzstplot? For the extra interactive feature, better simulation, even if the backend of gnuplot or OpenGL from Hamzstplot is already good enough, the features in Hamzstlab Mathematics that is coming from ImGUI, Implot, Implot3D, Imnodes can reach to almost all branches of science and engineering, help them to design and make prototype. Box2D is using ImGUI and it helps me personally to create some basic Physics simulation from an undergraduate book "Fundamental of Physics."

You can design any kind of electric circuit here with all the computation necessary too, you can simulate how a mechanical system works, how leverage works, how a human organ works, how a nuclear fission works, and you can even do gene sequence simulation better here, they are all in my head now, but one day we will add the source code and API so people can really create what I am talking now.

In the end, there is no perfect software, one software or one C++ library complement the other. That is life. Just like marriage, since nobody is perfect, just be honest, no lie no cheating, accept your partner for whoever she is, with no extramarital affair / lie / cheating, then you both will be happy and reach your goal together, with a bonus become power couple too.

I. BUILD AND INSTALL HAMZSTLAB MATHEMATICS IN GFREYA OS

We are using GFreya OS as it is a custom Linux-based Operating System, your distro your rules. The dependencies are:

1. ImGUI version 1.92.0 WIP
2. Implot version 0.17
3. Implot-3D version 0.3 WIP
4. Imnodes

There is no need to download all of the above one by one anymore, because they are already blended into one in Hamzstlab Mathematics.

To download Hamzstlab Mathematics, open terminal / xterm then type:
git clone https://github.com/glanzkaizer/Hamzstlab-Mathematics.git

Enter the directory then type:

```
cd dependencies  
  
cp -r cxxopts.hpp /usr/include
```

Code 5: Move an include file / header

For the **implot-demos** we need a few compiling first, from the main directory open the terminal then type:

```
cd implot-demos  
  
mkdir build  
cd build  
cmake ..  
make  
  
cp -r libimnodes.a libimplot.a libapp.a libimgui.a ../../dependencies/
```

Code 6: Create implot-demos related static library

The steps above are used to create the static library to run some demos from **implot-demos** that are more complex than the normal implot demo.

Alternative way besides static library:

You don't really need to create all the static libraries, considering if the dependencies are updated and you need a new feature. We think it is better to take the source files instead, e.g. if you take a look at our example:

Hamzstlab-Mathematics/examples/ImNodes Test/

It is added on June 18th, 2025. We don't use **libimnodes.a**, instead we put the source files for **Imnodes** (imnodes.cpp, imnodes.h, imnodes_internal.h) on the parent directory **Hamzstlab-Mathematics/** and adjust the Makefile in that example to process the source codes of Imnodes into

object files so the example that needs **Imnodes** feature will be able to be built as executable and run.

Now, you are set to go, in the **Hamzstlab-Mathematics/examples/** directory there are tons of examples that you can just build with Makefile, easy.

For example, from the main directory **Hamzstlab-Mathematics/** open terminal

```
cd examples

cd 3D Test

make

./main
```

Code 7: *Build an example*

This will run the 3D test. You can check each examples source code and learn how to modify according to your needs.

All files, examples codes and source codes used in this books can be found in this repository:
<https://github.com/glanzkaiser/Hamzstlab-Mathematics>

II. EXAMPLE OF CREATING AN INTERACTIVE PLOT WITH HAMZSTLAB-MATHEMATICS IN GFREYA OS

First you have to already clone or download the repository, then open terminal at the main working directory **Hamzstlab-Mathematics** and type:

```
cd examples

cd First Order Differential Equation

make

./main
```

Code 8: *Create dynamic library*

This will be the result:

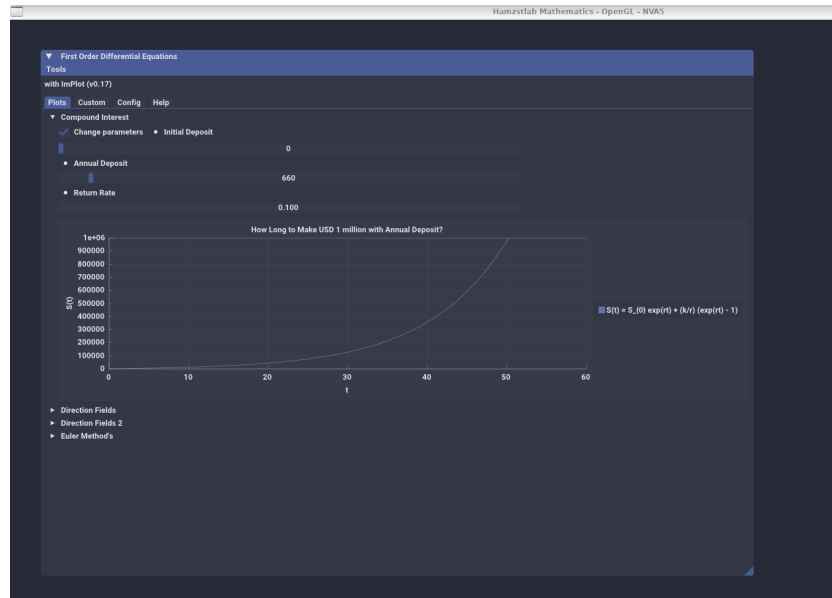


Figure 3.1: The interactive plot of $S(t)$ with Hamzstlab-Mathematics you can change the parameters and the plot will adjust with the parameters that you just input.

Now, we are going to explain the code on how to make this example works.

```
#include "App.h"

struct ImPlotDemo : App {
    using App::App;
    void Update() override {
        ImPlot::ShowFirstOrderDEWindow();
    }
};

int main(int argc, char const *argv[])
{
    ImPlotDemo app("Hamzstlab Mathematics", 1920, 1080, argc, argv);
    app.Run();

    return 0;
}
```

Code 9: *Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp*

This **main.cpp** is pointing toward **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**, so if you want to modify the example, you just need to edit **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**.

The **ImPlot::ShowFirstOrderDEWindow();** needs to be an **IMPLOT_API**, and this can be done in **Hamzstlab-Mathematics/implot.h**, so you need to add a line like this there

```
...  
  
IMPLICIT_API void ShowFirstOrderDEWindow(bool* p_open = nullptr);  
  
...
```

Code 10: *Hamzstlab-Mathematics/implicit.h*

For example, from hundred lines of codes you can check the code here to create a plot of Compound Interest, if you scroll down again you will know that we are showing this **Demo_CompoundInterest()** under a new tab that can be opened by the user.

There might be hundreds to thousands of lines of codes, but if you are rigorous and persistence you will get used to it and able to create something from your imagination.

```
...  
...  
  
void Demo_CompoundInterest() {  
    static float xs1[1001], ys1[1001];  
    static int S0 = 0;  
    static float r = 0.08;  
    static int k = 2000;  
    for (int i = 0; i < 1001; ++i) {  
        xs1[i] = i;  
        ys1[i] = S0 * exp(r*i) + (k/r)*(exp(r*i) - 1) ;  
    }  
  
    static bool range = false;  
    ImGui::Checkbox("Change parameters", &range);  
  
    if (range) {  
        ImGui::SameLine();  
        ImGui::SetNextItemWidth(200);  
        ImGui::BulletText("Initial Deposit");  
        ImGui::SliderInt("##Initial Deposit", &S0, 0, 10000);  
        ImGui::SetNextItemWidth(200);  
        ImGui::BulletText("Annual Deposit");  
        ImGui::SliderInt("##Annual Deposit", &k, 1, 10000);  
        ImGui::SetNextItemWidth(200);  
        ImGui::BulletText("Return Rate");  
        ImGui::DragFloat("##Return rate", &r, 0.01f, 0.2f);  
        ImGui::SetNextItemWidth(200);  
    }  
    if (ImGui::BeginPlot("How Long to Make USD 1 million with Annual  
        Deposit?")) {  
        ImGui::SetupAxes("t", "S(t)");  
        ImGui::SetupAxesLimits(0, 60, 0, 1000000);  
        ImGui::SetupLegend(ImGuiLocation_East,
```

```

        ImPlotLegendFlags_Outside);

        ImPlot::PlotLine("S(t) = S_{0} exp(rt) + (k/r) (exp(rt) - 1)
        ", xs1, ys1, 1001);
        ImPlot::EndPlot();
    }
}
...
...

void ShowFirstOrderDEWindow(bool* p_open) {
...

if (ImGui::BeginTabBar("ImPlotDemoTabs")) {
if (ImGui::BeginTabItem("Plots")) {
    DemoHeader("Compound Interest", Demo_CompoundInterest);

...
...
void ImPlot::ShowFirstOrderDEWindow(bool* p_open) {}

#endif

```

Code 11: *Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp*

Last, but not least, for the Makefile **Hamzstlab-Mathematics/examples/First Order Differential Equation/Makefile**, we need to add the SOURCES to be compiled that is pointing toward **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**

```

EXE = main
IMGUI_DIR = ../..
SOURCES = main.cpp
SOURCES += $(IMGUI_DIR)/imgui.cpp $(IMGUI_DIR)/imgui_demo.cpp $(IMGUI_DIR)/
imgui_draw.cpp $(IMGUI_DIR)/imgui_tables.cpp $(IMGUI_DIR)/imgui_widgets
.cpp
SOURCES += $(IMGUI_DIR)/backends/imgui_impl_glfw.cpp $(IMGUI_DIR)/backends/
imgui_impl_opengl3.cpp
SOURCES += $(IMGUI_DIR)/implplot.cpp $(IMGUI_DIR)/implplot_items.cpp
SOURCES += $(IMGUI_DIR)/hamzstlab_firstorderdifferentialequations.cpp

OBJS = $(addsuffix .o, $(basename $(notdir $(SOURCES))))
UNAME_S := $(shell uname -s)
LINUX_GL_LIBS = -lGL -lglfw

CXXFLAGS = -std=c++11 -I$(IMGUI_DIR) -I$(IMGUI_DIR)/backends
CXXFLAGS += -g -Wall -Wformat
LIBS = ../dependencies/glad.c -L../dependencies/ -lapp -limgui -
limnodes -limplot

```



```
...  
...
```

Code 12: *Hamzstlab-Mathematics/examples/First Order Differential Equation/Makefile*

Chapter 4

Intermediate Calculus

"Ma n'atu sole, che bello oje ne. O Sole Mio, sta fronte a te" - O Sole Mio

W^E are going to learn how to plot, simulate and compute with either SymIntegration, Hamzstplot, or Hamzstlab Mathematics for this chapter, we are following the Calculus book [7] from chapter 9 to 14 for the theorems, problems and definition. The chapter 1-8 are covered in Lasthrim Projection book [6] and that book is still using Python and JULIA.

I. INFINITE SERIES

II. CONICS AND POLAR COORDINATES

- General Theory of n th Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

III. GEOMETRY IN SPACE AND VECTORS

- [Vector-Valued Functions and Curvilinear Motion](#)

[\[H*\]](#) The first
[\[H*\]](#)

- [Surfaces in Three-Space](#)

[\[H*\]](#) The first
[\[H*\]](#)

IV. DERIVATIVES FOR FUNCTIONS OF TWO OR MORE VARIABLES

- Functions of Two or More Variables

[H*] The first
[H*]

- Partial Derivatives

[H*] The first
[H*]

V. MULTIPLE INTEGRALS

- Double Integrals over Rectangles

[H*] The first
[H*]

- Iterated Integrals

[H*] The first
[H*]

VI. VECTOR CALCULUS

- Vector Fields

[H*] The first
[H*]

- Line Integrals

[H*] The first
[H*]

Chapter 5

Differential Equations

"Everyone's favorite fighting fairy godmother is still kicking, taking out Ether Strike hits for her godfather Odin. The heretofore unmatched fury of her scowl derives from her lack of lines in the main story" - Freya (Valkyrie Profile: Covenant of the Plume)

Differential equations are equations that contain derivatives. It is used to understand and to investigate problems involving the motion of fluids, the flow of current in electric circuits, the dissipation of heat in solid objects, the propagation and detection of seismic waves [1]. For example, the Newton's second law can be written as a differential equation

$$F = ma \quad (5.1)$$

$$F = m \frac{dv}{dt} = mv' \quad (5.2)$$

We think of v as a function of t , or we can also write it as $v(t)$.

For this chapter, we will follow the outline of [1] book and we start it on May 2025, we will take the measure how long it takes to finish this chapter that is equal to the book mentioned.

I. DIRECTION FIELDS

[H*] Direction fields are valuable tools in studying the solutions of differential equations of the form

$$\frac{dy}{dt} = f(t, y) \quad (5.3)$$

where f is a given function of two variables t and y , sometimes referred to as the rate function.

A direction field (slope field) is a mathematical object used to graphically represent solutions to a first-order differential equation. At each point in a direction field, a line segment appears whose slope is equal to the slope of a solution to the differential equation passing through that point.

A direction field can be constructed by evaluating f at each point of a rectangular grid. At each point of the grid, a short line segment is drawn whose slope is the value of f at that point. Thus each line segment is tangent to the graph of the solution passing through that point.

[H*] A direction field drawn on a fairly fine grid gives good picture of the overall behavior of solutions of a differential equation.

The construction of a direction field is often a useful step in the investigation of a differential equation.

[H*] **Example:**

Consider a population of field mice who inhabit a certain rural area. If we denote time by t and the mouse population by $p(t)$ with the time is measured in months, then the differential that represents the population growth of field mice that is also preyed on by owls can be expressed by the equation

$$\frac{dp}{dt} = 0.5p - 450$$

Observe that the predation term is measured in months as -450 , meaning that each month there is about 450 mouse being preyed by owls every month.

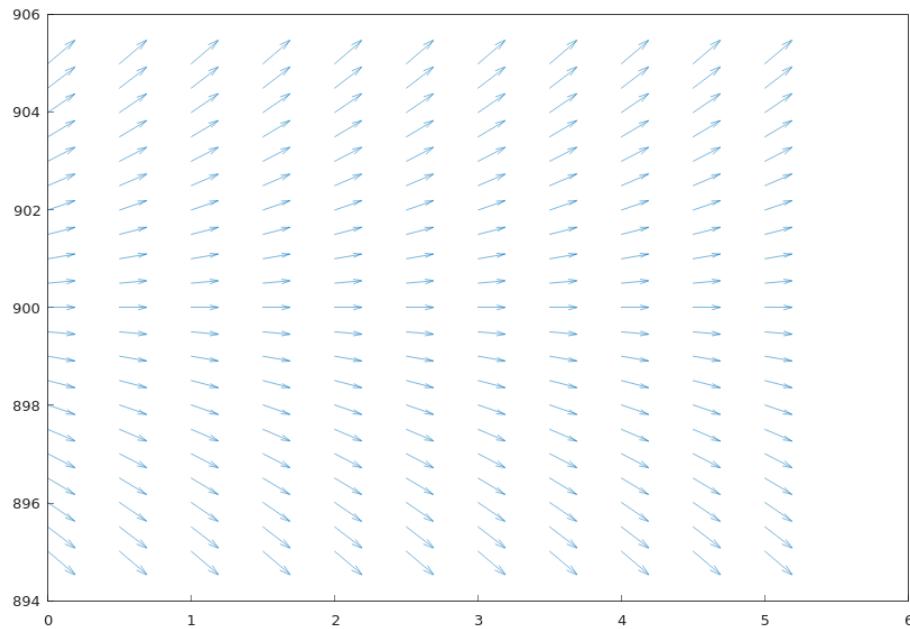


Figure 5.1: The direction field for $\frac{dp}{dt} = 0.5p - 450$, the x -axis represents the variable t as the time and the y -axis represents variable $p(t)$ as the population growth of field mice, we can see that if the rate of the reproduction for the field mice is greater than the rate of the mice being preyed on by owls then the population growth will keep increasing, and if the rate of the reproduction for the field mice is smaller than the rate of the mice being preyed on by owls then the population of field mice will keep decreasing.

The code to create the direction field above with Hamzstplot:

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;
```

```
int main() {
    using namespace hamzstplot;
    double dx, dy, magnitude;
    auto [x, y] = meshgrid(iota(0, 0.5, 5), iota(895, 0.5,
        905));

    vector_2d u = transform(x, y, [](double x, double y) {
        return 1 ; }); // dx
    vector_2d v = transform(x, y, [](double x, double y) {
        return 0.5*y - 450 ; }); // dy
    quiver(x, y, u, v);

    show();
    return 0;
}
```

Code 13: *Hamzstplot/Examples with Makefile/Plot Direction Fields Growth and Predation Rate/main.cpp*

For sufficiently large values of p it can be seen that $\frac{dp}{dt}$ is positive, so that solutions increase. On the other hand, if p is small, then $\frac{dp}{dt}$ is negative and solutions decrease.

The critical value of p that separates solutions that increase from those that decrease is the value of p for which $\frac{dp}{dt}$ is zero.

By setting $\frac{dp}{dt}$ equals zero and then solving for p , we find the equilibrium solution $p(t) = 900$ for which the growth term and the predation term are exactly balanced.

II. SOLVING A SIMPLE DIFFERENTIAL EQUATION

[H*] Consider we have this differential equation

$$\frac{dp}{dt} = 0.5p - 450$$

If we want to obtain the solution $p(t)$ we will have to perform some algebraic trick or manipulation then integrating them with respect to the correct variable.

$$\begin{aligned}
 \frac{dp}{dt} &= 0.5p - 450 \\
 &= \frac{p - 900}{2} \\
 \frac{dp/dt}{p - 900} &= \frac{1}{2} \\
 \frac{dp}{p - 900} &= \frac{dt}{2} \\
 \int \frac{dp}{p - 900} &= \int \frac{dt}{2} \\
 \ln |p - 900| &= \frac{t}{2} + C \\
 |p - 900| &= e^{\frac{t}{2} + C} \\
 |p - 900| &= e^{\frac{t}{2}} e^C \\
 p - 900 &= \pm e^{\frac{t}{2}} e^C \\
 p &= 900 + ce^{\frac{t}{2}}
 \end{aligned}$$

where $c = \pm e^C$ is also an arbitrary (nonzero) constant. Note that the constant function $p = 900$ is also a solution where $c = 0$.

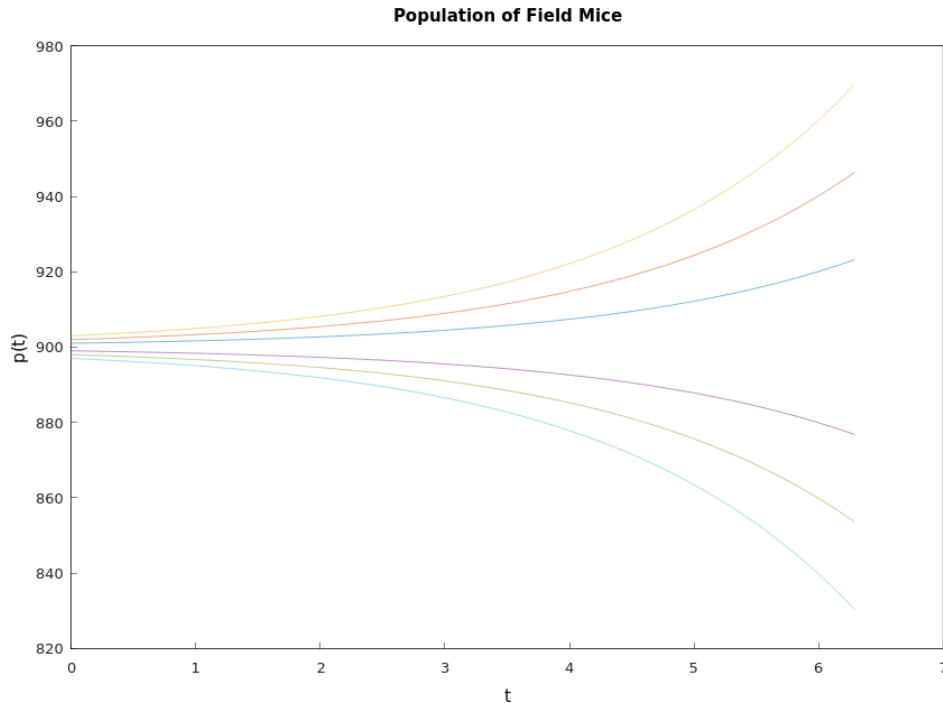


Figure 5.2: The graphs shows the solutions for $\frac{dp}{dt} = 0.5p - 450$ with several values of c .

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;

int main() {
    using namespace hamzstplot;

    std::vector<double> t = linspace(0, 2 * pi);
    std::vector<double> y1 = transform(t, [](auto t) {
        return 900 + 1*exp(0.5*t); });
    std::vector<double> y2 = transform(t, [](auto t) {
        return 900 + 2*exp(0.5*t); });
    std::vector<double> y3 = transform(t, [](auto t) {
        return 900 + 3*exp(0.5*t); });
    std::vector<double> y4 = transform(t, [](auto t) {
        return 900 + -1*exp(0.5*t); });
    std::vector<double> y5 = transform(t, [](auto t) {
        return 900 + -2*exp(0.5*t); });
    std::vector<double> y6 = transform(t, [](auto t) {
        return 900 + -3*exp(0.5*t); });

    plot(t, y1, t, y2, t, y3, t, y4, "-", t, y5, "-", t, y6,
        "-");
    title("Population of Field Mice");
    xlabel("t");
    ylabel("p(t)");
    show();
    return 0;
}
```

Code 14: *Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Solutions Curves/main.cpp*

III. CLASSIFICATION OF DIFFERENTIAL EQUATIONS

[H*] One important classification is based on whether the unknown function depends on a single independent variable or on several independent variables.

If a function depends on a single independent variable then it is said to be an ordinary differential equation, this is the example:

$$L \frac{d^2 Q(t)}{dt^2} + R \frac{dQ(t)}{dt} + \frac{1}{C} Q(t) = E(t) \quad (5.4)$$

with $Q(t)$ represents the charge on a capacitor in a circuit with capacitance C , resistance R , and inductance L .

If a function depends on more than one independent variable then it is called a partial differential equation, this is the example:

$$\alpha^2 \frac{\partial^2 u(x, t)}{\partial x^2} = \frac{\partial u(x, t)}{\partial t} \quad (5.5)$$

it is called the heat conduction equation with *alpha* as the physical constant. Another example is:

$$\alpha^2 \frac{\partial^2 u(x, t)}{\partial x^2} = \frac{\partial^2 u(x, t)}{\partial t^2} \quad (5.6)$$

it is called the wave equation with α as the physical constant.

[H*] System of differential equations usually have two or more unknown functions. For example, the Lotka-Volterra equations:

$$\begin{aligned} \frac{dx}{dt} &= ax - \alpha xy \\ \frac{dy}{dt} &= -cy + \gamma xy \end{aligned} \quad (5.7)$$

where $x(t)$ and $y(t)$ are the respective populations of the prey and predator species. The constants a, α, c , and γ are based on empirical observations and depend on the particular species being studied.

[H*] The order of a differential equation is the order of the highest derivative that appears in the equation.

More generally, the equation

$$F[t, u(t), u'(t), \dots, u^{(n)}(t)] = 0 \quad (5.8)$$

is an ordinary differential equation of the n th order.

Another one,

$$y''' + 2e^t y'' + y y' = t^3$$

is a third order differential equation for $y = u(t)$.

[H*] The ordinary differential equation

$$F(t, y, y', \dots, y^{(n)}) = 0$$

is said to be linear if F is a linear function of the variables $y, y', \dots, y^{(n)}$; a similar definition applies to partial differential equations.

Thus, the general linear ordinary differential equation of order n is

$$a_0(t)y^{(n)} + a_1(t)y^{(n-1)} + \dots + a_n(t)y = g(t) \quad (5.9)$$

A simple physical problem that leads to a nonlinear differential equation is the oscillating pendulum. The angle θ that an oscillating pendulum of length L makes with the vertical direction satisfies the equation

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \sin \theta = 0 \quad (5.10)$$

the term involving $\sin \theta$ makes that equation to be nonlinear.

IV. FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS

• Linear Equations; Method of Integrating Factors

[H*] Suppose we have this differential equations of first order

$$\frac{dy}{dt} = f(t, y)$$

where f is a given function of two variables. Any differentiable function $y = \phi(t)$ that satisfies this equation for all t in some interval is called a solution, and our object is to determine whether such functions exist.

For an arbitrary function f , there is no general method for solving the equation in terms of elementary functions.

[H*] We define the general first order linear equation in the standard form

$$\frac{dy}{dt} + p(t)y = g(t) \quad (5.11)$$

where p and g are given functions of the independent variable t .

To determine an appropriate integrating factor, we multiply Eq. (4.11) by an undetermined function $\mu(t)$, obtaining

$$\mu(t) \frac{dy}{dt} + p(t)\mu(t)y = \mu(t)g(t) \quad (5.12)$$

We see the left side of the Eq. (4.12) is the derivative of the product $\mu(t)y$, provided that $\mu(t)$ satisfies the equation

$$\frac{d\mu(t)}{dt} = p(t)\mu(t) \quad (5.13)$$

If we assume temporarily that $\mu(t)$ is positive, then we have

$$\begin{aligned} \frac{d\mu(t)/dt}{\mu(t)} &= p(t) \\ \ln |\mu(t)| &= \int p(t) dt + k \end{aligned}$$

By choosing the arbitrary constant k to be zero, we obtain the simplest possible function for μ , namely,

$$\mu(t) = e^{\int p(t) dt} \quad (5.14)$$

Note that $\mu(t)$ is positive for all t , as we assumed. Returning to Eq. (4.12), we have

$$\frac{d}{dt} [\mu(t)y] = \mu(t)g(t) \quad (5.15)$$

Hence

$$\mu(t)y = \int \mu(t)g(t) dt + c \quad (5.16)$$

where c is an arbitrary constant.

The general solution of Eq. (4.11) is

$$y = \frac{1}{\mu(t)} \left[\int_{t_0}^t \mu(s)g(s) ds + c \right] \quad (5.17)$$

where t_0 is some convenient lower limit of integration.

[H*] Example:

Solve the initial value problem

$$\begin{aligned} ty' + 2y &= 4t^2 \\ y(1) &= 2 \end{aligned}$$

Solution:

In order to determine $p(t)$ and $g(t)$ correctly, we must first rewrite the equation in the standard form of Eq. (4.11). Thus we have

$$y' + (2/t)y = 4t \quad (5.18)$$

so

$$\begin{aligned} p(t) &= \frac{2}{t} \\ g(t) &= 4t \end{aligned}$$

Now, we first compute the integrating factor $\mu(t)$:

$$\begin{aligned} \mu(t) &= e^{\int \frac{2}{t} dt} \\ &= e^{2 \ln |t|} \\ &= t^2 \end{aligned}$$

Following the formula of Eq. (4.17), we obtain

$$\begin{aligned} y &= \frac{1}{\mu(t)} \left[\int_{t_0}^t \mu(s)g(s) ds + c \right] \\ &= \frac{1}{t^2} \left[\int_0^t s^2(4s) ds \right] \\ &= \frac{1}{t^2} \left[\int_0^t 4s^3 ds \right] \\ &= \frac{1}{t^2} \left[t^4 + c \right] \\ y &= t^2 + \frac{c}{t^2} \end{aligned}$$

$y(t) = t^2 + \frac{c}{t^2}$ is the general solution of $ty' + 2y = 4t^2$.

To obtain the value for c , we will input the the initial condition value $y(1) = 2$ to the general solution

$$\begin{aligned} y(1) &= 2 \\ 2 &= 1^2 + \frac{c}{1^2} \\ c &= 1 \end{aligned}$$

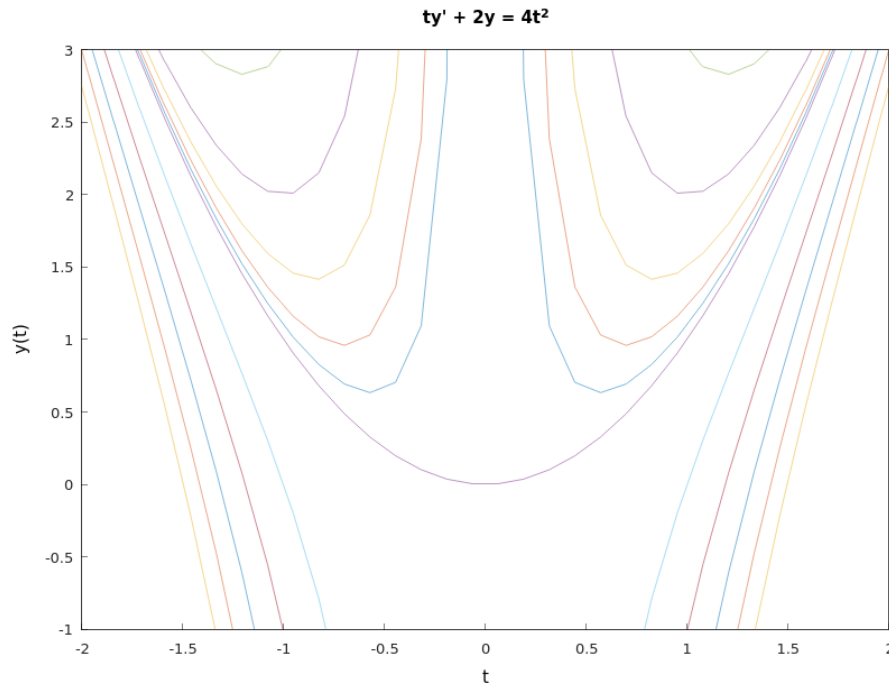


Figure 5.3: The graphs shows the solutions for $ty' + 2y = 4t^2$.

The solution becomes unbounded and is asymptotic to the positive y -axis as $t \rightarrow 0$ from the right. This is the effect of the infinite discontinuity in the coefficient $p(t)$ at the origin.

The function $y = t^2 + \frac{1}{t^2}$ for $t < 0$ is not part of the solution of this initial value problem. The solution fails to exist for some values of t . This is due to the infinite discontinuity in $p(t)$ at $t = 0$, which restricts the solution to the interval $0 < t < \infty$.

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;

int main() {
    using namespace hamzstplot;

    std::vector<double> t = linspace(-2 * pi, 2 * pi);
    std::vector<double> y1 = transform(t, [](auto t) { return
        pow(t,2) + 0.1/(pow(t,2)); });
    std::vector<double> y2 = transform(t, [](auto t) { return
        pow(t,2) + 0.23/(pow(t,2)); });
    std::vector<double> y3 = transform(t, [](auto t) { return
        pow(t,2) + 0.5/(pow(t,2)); });
    std::vector<double> y4 = transform(t, [](auto t) { return
        pow(t,2) + 1/(pow(t,2)); });
    std::vector<double> y5 = transform(t, [](auto t) { return
```

```

        pow(t,2) + 2/(pow(t,2)); });
std::vector<double> y6 = transform(t, [](auto t) { return
    pow(t,2) + -1/(pow(t,2)); });
std::vector<double> y7 = transform(t, [](auto t) { return
    pow(t,2) + -2/(pow(t,2)); });
std::vector<double> y8 = transform(t, [](auto t) { return
    pow(t,2) + -3/(pow(t,2)); });
std::vector<double> y9 = transform(t, [](auto t) { return
    pow(t,2) + -4/(pow(t,2)); });
std::vector<double> y10 = transform(t, [](auto t) { return
    pow(t,2) + -5/(pow(t,2)); });
std::vector<double> y11 = transform(t, [](auto t) { return
    pow(t,2); });

plot(t, y1, t, y2, t, y3, t, y4, "-", t, y5, "-", t, y6, "-",
    t, y7, "-", t, y8, "-", t, y9, "-", t, y10, "-", t,
    y11, "-");
axis({-2, 2, -1, 3});
title("ty' + 2y = 4t^2");
xlabel("t");
ylabel("y(t)");
show();
return 0;
}

```

Code 15: *Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Solutions Curves for Method of Integrating Factors/main.cpp*

- **Separable Equations**

[H*] The general first order equation is

$$\frac{dy}{dx} = f(x, y) \quad (5.19)$$

By setting $M(x, y) = -f(x, y)$ and $N(x, y) = 1$ we will have

$$M(x, y) + N(x, y) \frac{dy}{dx} = 0 \quad (5.20)$$

If it happens that M is a function of x only and N is a function of y only, then it becomes

$$M(x) + N(y) \frac{dy}{dx} = 0 \quad (5.21)$$

Such an equation is said to be separable, because if it is written in the differential form

$$M(x) dx + N(y) dy = 0 \quad (5.22)$$

The differential form (4.22) is also more symmetric and tends to suppress the distinction between independent and dependent variables.

A separable equation can be solved by integrating the functions M and N .

[H*] Example:

Show that the equation

$$\frac{dy}{dx} = \frac{x^2}{1-y^2}$$

is separable, and then find an equation for its integral curves.

Solution:

If we write the equation as

$$-x^2 + (1-y^2)\frac{dy}{dx} = 0$$

then it is separable. Recall from calculus that if y is a function of x , then by the chain rule

$$\frac{d}{dx}f(y) = \frac{d}{dy}f(y)\frac{dy}{dx} = f'(y)\frac{dy}{dx}$$

If $f(y) = y - \frac{y^3}{3}$, then

$$\frac{d}{dx}\left(y - \frac{y^3}{3}\right) = (1-y^2)\frac{dy}{dx}$$

Thus,

$$\begin{aligned} -x^2 + (1-y^2)\frac{dy}{dx} &= 0 \\ (1-y^2)dy &= x^2 dx \\ y - \frac{y^3}{3} &= \frac{x^3}{3} + c \\ -x^3 + 3y - y^3 &= c \end{aligned}$$

Any differentiable function $y = \phi(x)$ that satisfies $-x^3 + 3y - y^3 = c$ is a solution of $\frac{dy}{dx} = \frac{x^2}{1-y^2}$.

An equation of the integral curve passing through a particular point (x_0, y_0) can be found by substituting x_0 and y_0 for x and y , respectively, in $-x^3 + 3y - y^3 = c$ and determining the corresponding value of c .

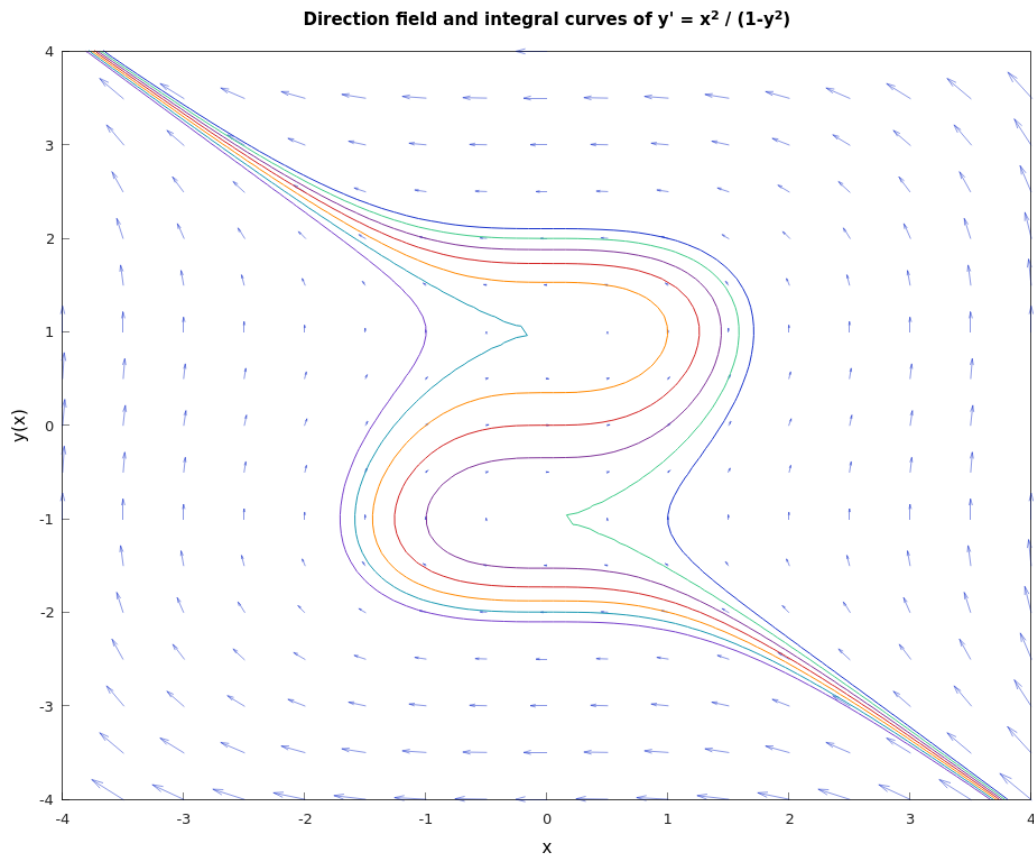


Figure 5.4: The graphs shows the direction field and integral curves of $y' = \frac{x^2}{1-y^2}$.

```
#include <cmath>
#include <hamzstplot/hamzstplot.h>

using namespace std;

int main() {
    using namespace hamzstplot;

    vector_2d newcolors = {{0.83, 0.14, 0.14},
                           {1.00, 0.54, 0.00},
                           {0.10, 0.6, 0.70},
                           {0.47, 0.25, 0.80},
                           {0.50, 0.2, 0.60},
                           {0.25, 0.80, 0.54}};
    colororder(newcolors);

    auto [x, y] = meshgrid(iota(-4, 0.5, 5), iota(-4, 0.5, 4))
        ;
    vector_2d u = transform(x, y, [] (double x, double y) {
```

```

        return 1-pow(y,2) ; }); // dx
vector_2d v = transform(x, y, [](double x, double y) {
    return pow(x,2) ; }); // dy

quiver(x, y, u, v);
hold(on);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) ; }, "-")->line_width(1);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) - 1; }, "-")->line_width(1);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) - 2; }, "-")->line_width(1);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) - 3; }, "-")->line_width(1);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) + 1; }, "-")->line_width(1);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) + 2; }, "-")->line_width(1);
fimplicit([(double x, double y) { return -pow(x, 3) + 3*y
    - pow(y, 3) + 3; }, "-")->line_width(1);
hold(off);

axis({-4, 4, -4, 4});
title("Integral curves of  $y' = x^2 / (1-y^2)$ ");
xlabel("x");
ylabel("y(x)");
show();
return 0;

}

```

Code 16: *Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Direction Field and Solutions Curves for Separable Equations/main.cpp*

The coding here is using **fimplicit** since we are inputting $-x^3 + 3y - y^3 - c = 0$ as the function to be drawn, it is the implicit function. We also draw the direction field so we are using **hold(on)** then closing it with **hold(off)**.

- **Modeling with First Order Equations**

[H*] Differential equations are of interest to nonmathematicians primarily because of the possibility of using them to investigate a wide variety of problems in the physical, biological, and social sciences.

One reason for this is that mathematical models and their solutions lead to equations relating the variables and parameters in the problem. These equations often enable you to make predictions about how the natural process will behave in various circumstances.

Mathematical modeling and experiment or observation are both critically important

and have somewhat complementary roles in scientific investigations. Mathematical models are validated by comparison of their predictions with experimental results.

[H*] Three identifiable steps that are always present in the process of mathematical modeling:

1. Construction of the Model.
2. Analysis of the Model.
3. Comparison with Experiment or Observation.

[H*] **Compound Interest**

Suppose that a sum of money is deposited in a bank that pays interest at an annual rate r . The value $S(t)$ of the investment at any time t depends on the frequency with which interest is compounded as well as on the interest rate.

Financial institutions have various policies concerning compounding: some compound monthly, some weekly, some even daily.

If we assume that compounding takes place continuously, then we can set up a simple initial value problem that describes the growth of the investment.

The rate of change of the value of the investment is $\frac{dS}{dt}$, and this quantity is equal to the rate at which interest accrues, which is the interest rate r times the current value of the investment $S(t)$. Thus

$$\frac{dS}{dt} = rS \quad (5.23)$$

is the differential equation that governs the process. Suppose that we also know the value of the investment at some particular time, say,

$$S(0) = S_0 \quad (5.24)$$

Then the solution of the initial value problem at Eq. (4.23) and Eq. (4.24) gives the balance $S(t)$ in the account at any time t . This initial value problem is readily solved, since the differential equation at Eq. (4.23) is both linear and separable. Consequently, by solving Eqs. (4.23) and (4.24), we find that

$$S(t) = S_0 e^{rt} \quad (5.25)$$

Thus a bank account with continuously compounding interest grows exponentially.

Let us suppose that there may be deposits or withdrawals in addition to the accrual of interest, dividends, or capital gains. If we assume that the deposits or withdrawals take place at a constant rate k , then Eq. (4.23) is replaced by

$$\frac{dS}{dt} = rS + k \quad (5.26)$$

where k is positive for deposits and negative for withdrawals.

Eq. (4.26) is linear with the integrating factor e^{-rt} , so its general solution is

$$S(t) = ce^{rt} - \frac{k}{r} \quad (5.27)$$

where c is an arbitrary constant. To satisfy the initial condition Eq. (4.24), we must choose $c = S_0 + \frac{k}{r}$. Thus the solution of the initial value problem is

$$S(t) = S_0 e^{rt} + \left(\frac{k}{r}\right) (e^{rt} - 1) \quad (5.28)$$

The first term in the expression (4.28) is the part of $S(t)$ that is due to the return accumulated on the initial amount S_0 , and the second term is the part that is due to the deposit or withdrawal rate k .

The advantage of stating the problem in this general way without specific values for S_0 , r , or k lies in the generality of the resulting formula (4.28) for $S(t)$. With this formula we can readily compare the results of different investment programs or different rates of return.

[H*] Example:

Suppose that one person opens an individual retirement account (IRA) at age 25 and makes annual investments of USD 2000 thereafter in a continuous manner. Assuming a rate of return of 8%, what will be the balance in the IRA at age 65?

Solution:

We have

$$\begin{aligned} S_0 &= 0 \\ r &= 0.08 \\ k &= 2000 \end{aligned}$$

and we wish to determine $S(40)$. From Eq. (4.28) we have

$$\begin{aligned} S(40) &= (0)e^{(0.08)(40)} + \left(\frac{2000}{0.08}\right) (e^{(0.08)(40)} - 1) \\ &= (25,000)(e^{3.2} - 1) \\ &= 588,313 \end{aligned}$$

It is interesting to note that the total amount invested is USD 80,000 ($40 \times 2,000$), so the remaining amount of USD 508,313 results from the accumulated return on the investment. The balance after 40 years is also fairly sensitive to the assumed rate.

For the C++ code, we are using Hamzstlab Mathematics for the first time as the introduction, this is to complement Hamzstplot beautifully. Hamzstlab Mathematics is using ImGUI and ImPlot so it can make the interactive GUI like in this example.

We can of course plot $S(t)$ easily with various rate with Hamzstplot, but we want a slider that can change the plot at an instant, that is why we are using Hamzstlab Mathematics, it can do the job better and faster.

We create a few examples for the first order ordinary differential equations section for Hamzstlab Mathematics. The code to create this is longer, not only one CPP file, but more than one, and it is a lot different than Hamzstplot, a lot of learning, you have to

read a lot for the codes, read about ImGui, Implot, steeper curve of learning, but it will worth your time since it can sustain not only decades but till another 100 years for the durability.

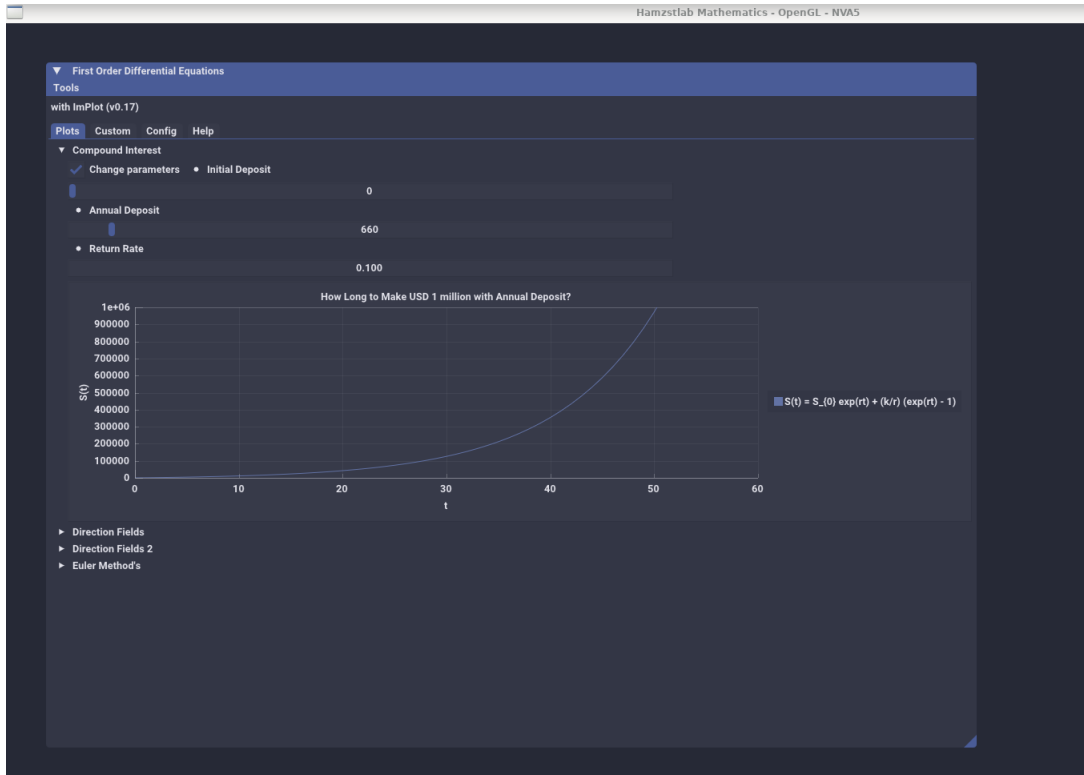


Figure 5.5: The interactive plot of $S(t)$ with Hamzstlab-Mathematics you can change the parameters and the plot will adjust with the parameters that you just input (Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp).

```
...
...

void Demo_CompoundInterest() {
    static float xs1[1001], ys1[1001];
    static int S0 = 0;
    static float r = 0.08;
    static int k = 2000;
    for (int i = 0; i < 1001; ++i) {
        xs1[i] = i;
        ys1[i] = S0 * exp(r*i) + (k/r)*(exp(r*i) - 1) ;
    }

    static bool range = false;
    ImGui::Checkbox("Change parameters", &range);
}
```



```
if (range) {
    ImGui::SameLine();
    ImGui::SetNextItemWidth(200);
    ImGui::BulletText("Initial Deposit");
    ImGui::SliderInt("##Initial Deposit", &S0, 0, 10000);
    ImGui::SetNextItemWidth(200);
    ImGui::BulletText("Annual Deposit");
    ImGui::SliderInt("##Annual Deposit", &k, 1, 10000);
    ImGui::SetNextItemWidth(200);
    ImGui::BulletText("Return Rate");
    ImGui::DragFloat("##Return rate", &r, 0.01f, 0.2f);
    ImGui::SetNextItemWidth(200);
}
if (ImGui::BeginPlot("How Long to Make USD 1 million with Annual
    Deposit?")) {
    ImGuiPlot::SetupAxes("t", "S(t)");
    ImGuiPlot::SetupAxesLimits(0, 60, 0, 1000000);
    ImGuiPlot::SetupLegend(ImGuiPlotLocation_East, ImGuiPlotLegendFlags_Outside
        );

    ImGuiPlot::PlotLine("S(t) = S_{0} exp(rt) + (k/r) (exp(rt) - 1)",
        xs1, ys1, 1001);
    ImGuiPlot::EndPlot();
}
}
...
...
```

Code 17: *Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp*

We only include the snippet of code to plot the $S(t)$, the full code can be seen at Hamzstlab Mathematics from the repository of Hamzstlab Mathematics at the **Hamzstlab-Mathematics/hamzstlab_firstorderdifferentialequations.cpp**, to run the example with Makefile you can go to this directory **Hamzstlab-Mathematics/examples/First Order Differential Equation**.

- [Differences Between Linear and Nonlinear Equations](#)

[H*] If you encounter an initial value problem in linear differential equation, you are guaranteed that a solution exists.

Theorem 5.1: Existence and Uniqueness of Solutions for Linear Differential Equation

If the functions p and g are continuous on an open interval $I : \alpha < t < \beta$ containing point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the differential equation

$$y' + p(t)y = g(t) \quad (5.29)$$

for each t in I , and that also satisfies the initial condition

$$y(t_0) = y_0 \quad (5.30)$$

where y_0 is an arbitrary prescribed initial value.

The theorem states that the solution exists throughout any interval I containing the initial point t_0 in which the coefficients p and g are continuous.

The solution can be discontinuous or fail to exist only at points where at least one of p and g is discontinuous.

[H*] For the nonlinear differential equations, we will use more general theorem

Theorem 5.2: Existence and Uniqueness of Solutions for Nonlinear Differential Equations

Let the functions f and $\frac{\partial f}{\partial y}$ be continuous in some rectangle $\alpha < t < \beta, \gamma < y < \delta$ containing the point (t_0, y_0) . Then, in some interval $t_0 - h < t < t_0 + h$ contained in $\alpha < t < \beta$, there is a unique solution $y = \phi(t)$ of the initial value problem

$$y' = f(t, y), \quad y(t_0) = y_0 \quad (5.31)$$

[H*] **Example:**

Solve the initial value problem

$$\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y - 1)}, \quad y(0) = -1$$

The differential equation is nonlinear, so

$$f(x, y) = \frac{3x^2 + 4x + 2}{2(y - 1)}$$

$$\frac{\partial f}{\partial y}(x, y) = -\frac{3x^2 + 4x + 2}{2(y - 1)^2}$$

Thus each of these functions is continuous everywhere except on the line $y = 1$. Consequently, a rectangle can be drawn about the initial point $(0, -1)$ in which both f and $\frac{\partial f}{\partial y}$ are continuous.

The theorem guarantes that the initial value problem has a unique solution in some interval about $x = 0$. However, even though the rectangle can be stretched infinitely far in both the positive and negative x directions, this does not necessarily mean that the solution exists for all x .

We can use separable equations technique

$$\begin{aligned}\frac{dy}{dx} &= \frac{3x^2 + 4x + 2}{2(y-1)} \\ 2(y-1)dy &= (3x^2 + 4x + 2)dx \\ y^2 - 2y &= x^3 + 2x^2 + 2x + c\end{aligned}$$

Now to solve the initial value problem we substitute $x = 0$ and $y = -1$ so we will obtain $c = 3$, thus

$$\begin{aligned}y^2 - 2y - (x^3 + 2x^2 + 2x + 3) &= 0 \\ y &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(-(x^3 + 2x^2 + 2x + 3))}}{2(1)} \\ y &= \frac{2 \pm \sqrt{4 + (4x^3 + 8x^2 + 8x + 12)}}{2} \\ y &= 1 \pm \sqrt{x^3 + 2x^2 + 2x + 4}\end{aligned}$$

To choose the plus or minus sign we need to check back again with the initial condition.

$$\begin{aligned}y(0) &= 1 - \sqrt{(0)^3 + 2(0)^2 + 2(0) + 4} \\ y(0) &= -1\end{aligned}$$

so we choose the minus sign for the solution. The initial value problem of $\frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)}$, $y(0) = -1$ has the solution of

$$y(x) = 1 - \sqrt{x^3 + 2x^2 + 2x + 4} \tag{5.32}$$

We must find the interval in which the quantity under the radical is positive (under the square root), since if it is negative then it will return complex number. The only real zero of this expression is $x = -2$, so the desired interval is $x > -2$, that means the solution exists only for $x > -2$.

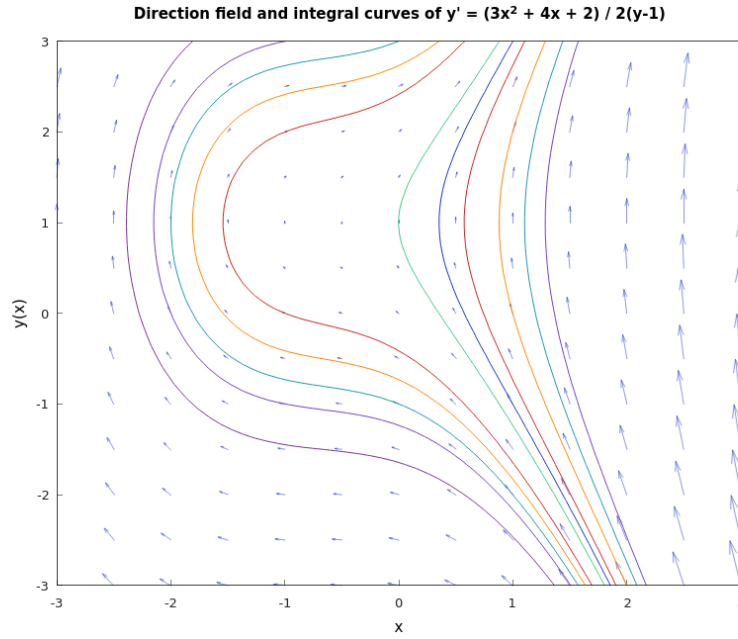


Figure 5.6: The direction field and integral curves of $\frac{dy}{dx} = \frac{3x^2+4x+2}{2(y-1)}$ (*Hamzstplot/Examples with Makefile/Plot First Order Differential Equation Direction Field and Solutions Curves for Separable Equations Example 2/main.cpp*).

[H*] Bernoulli Equations

Sometimes it is possible to solve a nonlinear equation by making a change of the dependent variable that converts it into a linear equation. The most important such equation has the form

$$y' + p(t)y = q(t)y^n \quad (5.33)$$

it is called a Bernoulli equation after Jakob Bernoulli.

[H*] Example:

- Solve Bernoulli' equation when $n = 0$ and when $n = 1$.
- Show that if $n \neq 0, 1$, then the substitution $v = y^{1-n}$ reduces Bernoulli's equation to a linear equation. This method of solution was found by Leibniz in 1696.

Solution:

- For $n = 0$ we will have

$$y' + p(t)y = q(t)$$

it is linear, so the solution is

$$y = \frac{1}{\mu(t)} \left[\int_{t_0}^t \mu(s)q(s) ds + c \right] \quad (5.34)$$

with

$$\mu(t) = e^{\int p(t) dt} \quad (5.35)$$

For $n = 1$ we will have

$$y' + p(t)y = q(t)y$$

it is linear again, so the solution is

$$y = \frac{1}{\mu(t)} [c] \quad (5.36)$$

with

$$\mu(t) = e^{\int p(t)-q(t) dt} \quad (5.37)$$

(b) Let $v = y^{1-n}$, then

$$\begin{aligned} \frac{dv}{dt} &= (1-n)y^{-n} \frac{dy}{dt} \\ \frac{dy}{dt} &= \frac{1}{1-n} y^n \frac{dv}{dt} \end{aligned}$$

which makes sense when $n \neq 0, 1$. Substituting into the differential equation yields

$$\begin{aligned} \frac{y^n}{1-n} \frac{dv}{dt} + p(t)y &= q(t)y^n \\ v' + (1-n)p(t)y^{1-n} &= (1-n)q(t) \end{aligned} \quad (5.38)$$

Setting $v = y^{1-n}$ then yields a linear differential equation for v . The Eq. (4.38) can be solved with method of integrating factors, it can also be written like this

$$v' + (1-n)p(t)v = (1-n)q(t) \quad (5.39)$$

[H*] Example:

The given equation is Bernoulli equation. In each case solve it by using the substitution method found by Leibniz in 1696.

- (a) $y' = \epsilon y - \sigma y^3$, $\epsilon > 0$ and $\sigma > 0$. This equation occurs in the study of the stability of fluid flow.
- (b) $y' = (\Gamma \cos t + T)y - y^3$, where Γ and T are constants. This equation also occurs in the study of the stability of fluid flow.

Solution:

- (a) Since $n = 3$, set $v = y^{-2}$. It follows that

$$\begin{aligned} \frac{dv}{dt} &= -2y^{-3} \frac{dy}{dt} \\ \frac{dy}{dt} &= -\frac{y^3}{2} \frac{dv}{dt} \end{aligned}$$

Substituting into the differential equation yields

$$\begin{aligned} y' &= \epsilon y - \sigma y^3 \\ -\frac{y^3}{2} \frac{dv}{dt} &= \epsilon y - \sigma y^3 \\ -\frac{y^3}{2} \frac{dv}{dt} - \epsilon y &= -\sigma y^3 \\ \frac{dv}{dt} + 2\epsilon y^{-2} &= 2\sigma \\ v' + 2\epsilon v &= 2\sigma \end{aligned}$$

This differential equation $v' + 2\epsilon v = 2\sigma$ is linear, and can be written as

$$(v \cdot e^{2\epsilon t})' = 2\sigma e^{2\epsilon t} \quad (5.40)$$

with the integrating factor $\mu(t) = e^{2\epsilon t}$. The solution is given by

$$v(t) = \frac{\sigma}{\epsilon} + ce^{-2\epsilon t} \quad (5.41)$$

Converting back to the original dependent variable, we will have

$$y = \pm v^{-\frac{1}{2}}$$

$$y(t) = \pm \frac{1}{\sqrt{\frac{\sigma}{\epsilon} + ce^{-2\epsilon t}}} \quad (5.42)$$

(b) Since $n = 3$, set $v = y^{-2}$. It follows that

$$\frac{dv}{dt} = -2y^{-3} \frac{dy}{dt}$$

$$\frac{dy}{dt} = -\frac{y^3}{2} \frac{dv}{dt}$$

The differential equation is written as

$$y' = (\Gamma \cos t + T)y - y^3$$

$$-\frac{y^3}{2} \frac{dv}{dt} = (\Gamma \cos t + T)y - y^3$$

$$-\frac{y^3}{2} \frac{dv}{dt} - (\Gamma \cos t + T)y = -y^3$$

$$\frac{dv}{dt} + 2(\Gamma \cos t + T)y^{-2} = 2$$

$$v' + 2(\Gamma \cos t + T)v = 2$$

This ordinary differential equation is linear, with integrating factor

$$\mu(t) = e^{2(\Gamma \sin t + Tt)}$$

Thus, we will have

$$(v \cdot e^{2(\Gamma \sin t + Tt)})' = 2e^{2(\Gamma \sin t + Tt)} \quad (5.43)$$

Then we can find the solution with the method of integrating factor

$$v(t) = \frac{1}{\mu(t)} \left[\int_{t_0}^t 2e^{2(\Gamma \sin s + Ts)} ds + c \right] \quad (5.44)$$

the solution is given by

$$v(t) = \frac{1}{e^{2(\Gamma \sin t + Tt)}} \left[\int_{t_0}^t 2e^{2(\Gamma \sin s + Ts)} ds + c \right] \quad (5.45)$$

the integral $\int_{t_0}^t 2e^{2(\Gamma \sin s + Ts)} ds$ is not an easy one to be integrated, even with SymPy in 2025, so we will keep it that way. We can also write it in this form

$$v(t) = (e^{-2(\Gamma \sin t + Tt)}) \int_{t_0}^t 2e^{2\Gamma \sin s} e^{2Ts} ds + c(e^{-2(\Gamma \sin t + Tt)}) \quad (5.46)$$

- Autonomous Equations and Population Dynamics

[H*] An important class of first order equations consists of those in which the independent variable does not appear explicitly. Such equations are called autonomous and have the form

$$\frac{dy}{dt} = f(y) \quad (5.47)$$

Equation (4.47) is separable, but the main purpose of this section is to show how geometrical methods can be used to obtain important qualitative information directly from the differential equation, without solving the equation.

[H*] **Exponential Growth**

Let $y = \phi(t)$ be the population of the given species at time t . The simplest hypothesis concerning the variation of population is that the rate of change of y is proportional to the current value of y ; that is,

$$\frac{dy}{dt} = ry \quad (5.48)$$

where the constant of proportionality r is called the rate of growth or decline, depending on whether it is positive or negative. Here, we assume that $r > 0$, so the population is growing.

Solving Eq. (4.48) subject to the initial condition

$$y(0) = y_0 \quad (5.49)$$

we obtain

$$y = y_0 e^{rt} \quad (5.50)$$

Thus the mathematical model consisting of the initial value problem (4.48), (4.49) with $r > 0$ predicts that the population will grow exponentially for all time. However, it is clear that such ideal conditions cannot continue indefinitely; eventually, limitations on space, food supply, or other resources will reduce the growth rate and bring an end to uninhibited exponential growth.

[H*] **Logistic Growth**

Consider the differential equation that the growth rate actually depends on the population

$$\frac{dy}{dt} = h(y)y \quad (5.51)$$

We now want to choose $h(y)$ so that $h(y) \equiv r > 0$ when y is small, $h(y)$ decreases as y grows larger, and $h(y) < 0$ when y is sufficiently large. The simplest function that has these properties is

$$h(y) = r - ay$$

where a is also a positive constant. Using this function in Eq. (4.51), we obtain

$$\frac{dy}{dt} = (r - ay)y \quad (5.52)$$

Equation (4.52) is known as the Verhulst equation or the logistic equation. It is often convenient to write the logistic equation in the equivalent form

$$\frac{dy}{dt} = r \left(1 - \frac{y}{K}\right) y \quad (5.53)$$

where $K = r/a$. The constant r is called the intrinsic growth rate, that is, the growth rate in the absence of any limiting factors.

We first seek solutions of Eq. (4.53) of the simplest possible type, that is, constant functions.

For such solution $\frac{dy}{dt} = 0$ for all t , so any constant solution of Eq. (4.53) must satisfy the algebraic equation

$$r \left(1 - \frac{y}{K} \right) y = 0$$

Thus the constant solutions are $y = \phi_1(t) = 0$ and $y = \phi_2(t) = K$. These solutions are called equilibrium solutions of Eq. (4.53) because they correspond to no change or variation in the value of y as t increases.

In the same way, any equilibrium solutions of the more general Eq. (4.47) can be found by locating the roots of $f(y) = 0$. The zeros of $f(y)$ are also called critical points.

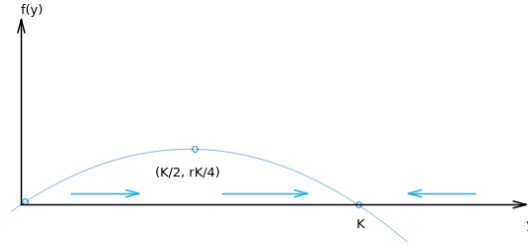


Figure 5.7: The equilibrium solution $f(y) = 0$ or $r(1 - y/K)y = 0$ to visualize other solutions of Eq. (4.53) (*Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Logistic Growth/main.cpp*).

The graph of $f(y)$ versus y above is a parabola with the intercepts are $(0,0)$ and $(K,0)$, corresponding to the critical points of Eq. (4.53), and the vertex of the parabola is $(K/2, rK/4)$.

Observe that $\frac{dy}{dt} > 0$ for $0 < y < K$; therefore, y is an increasing function of t when y is in this interval; this is indicated by the rightward-pointing arrows near the y -axis. Similarly, if $y > K$, then $\frac{dy}{dt} < 0$; hence y is decreasing, as indicated by the leftward-pointing arrow.

The y -axis is often called the phase line, and it is usually drawn with vertical line (rotating it 90 degree counterclockwise).

The dots at $y = 0$ and $y = K$ are the critical points or equilibrium solutions. The arrows again indicate that y is increasing whenever $0 < y < K$ and that y is decreasing whenever $y > K$.

Note that, if y is near zero or K , then the slope $f(y)$ is near zero, so the solution curves are relatively flat. They become steeper as the value of y leaves the neighborhood of zero or K .

To sketch the graphs of solutions of Eq. (4.53) in the ty -plane, we start with the

equilibrium solutions $y = 0$ and $y = K$; then we draw other curves that are increasing when $0 < y < K$, decreasing when $y > K$, and flatten out as y approaches either the values 0 or K . Thus, the graphs of solutions of Eq. (4.53) must have the general shape regardless of the values of r and K .

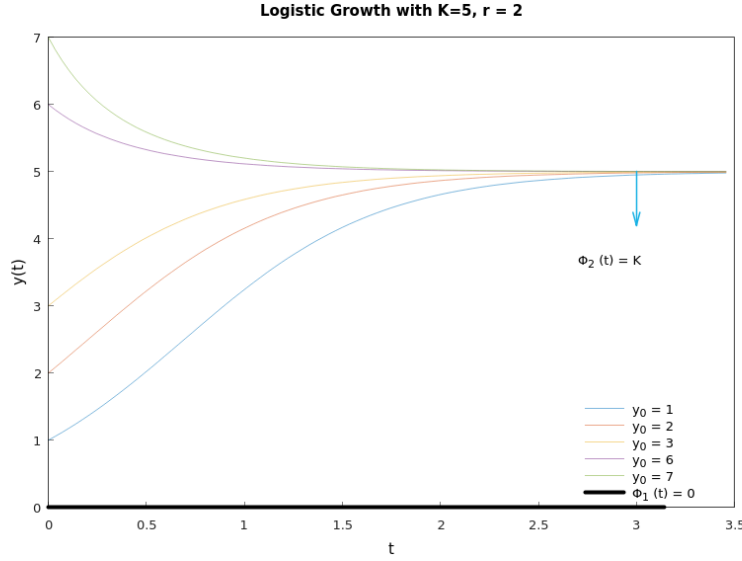


Figure 5.8: The solution curves for $y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}}$ to visualize all possible solutions of Eq. (4.53) with different y_0 (Hamzstplot/Examples with Makefile/Plot 2D Solution Curves for Logistic Growth/main.cpp).

It may seem that other solutions intersect the equilibrium solution at $y(t) = K$ in Figure (4.8), but it is not possible.

Based on the uniqueness part of Theorem 4.2, the fundamental existence and uniqueness theorem, states that only one solution can pass through a given point in the ty -plane. Thus, although other solutions may be asymptotic to the equilibrium solution as $t \rightarrow \infty$, they cannot intersect it at any finite time.

We can determine the concavity of the solution curves and the location of inflection points by finding $\frac{d^2 y}{dt^2}$. From the differential equation (4.47) we obtain (using the chain rule)

$$\frac{d^2 y}{dt^2} = \frac{d}{dt} \frac{dy}{dt} = \frac{d}{dt} f(y) = f'(y) \frac{dy}{dt} = f'(y) f(y) \quad (5.54)$$

The graph of y versus t is concave up when $y'' > 0$, that is, when f and f' have the same sign.

Similarly, it is concave down when $y'' < 0$, which occurs when f and f' have opposite signs. The signs of f and f' can be easily identified from the graph of $f(y)$ versus y . Inflection points may occur when $f'(y) = 0$.

In the case of Eq. (4.53), solutions are concave up for $0 < y < K/2$ where f is positive and increasing, so that both f and f' are positive. Solutions are also concave

up for $y > K$ where f is negative and decreasing (both f and f' are negative). For $K/2 < y < K$, solutions are concave down since here f is positive and decreasing, so f is positive but f' is negative. There is an inflection point whenever the graph of y versus t crosses the line $y = K/2$.

Finally, observe that K is the upper bound that is approached, but not exceeded, by growing populations starting below this value. Thus it is natural to refer to K as the saturation level, or the environmental carrying capacity, for the given species.

It reveals that solutions of the nonlinear equation (4.53) are strikingly different from those of the linear equation (4.47), at least for large values of t . Regardless of the value of K , that is, no matter how small the nonlinear term in Eq. (4.53), solutions of that equation approach a finite values as $t \rightarrow \infty$, whereas solutions of Eq. (4.47) grow (exponentially) without bound as $t \rightarrow \infty$. Thus, even a tiny nonlinear term in the differential equation (4.53) has a decisive effect on the solution for large t .

In many situations it is sufficient to have the qualitative information about a solution $y = \phi(t)$ of Eq. (4.53). This information was obtained entirely from the graph of $f(y)$ versus y , and without solving the differential equation (4.53). However, if we wish to know the value of the population at some particular time, then we must solve Eq. (4.53) subject to the initial condition (4.49). Provided that $y \neq 0$ and $y \neq K$, we can write Eq. (4.53) in the form

$$\begin{aligned} \frac{dy}{\left(1 - \frac{y}{K}\right)y} &= r dt \\ \left(\frac{1}{y} - \frac{1/K}{1 - y/K}\right) dy &= r dt \\ \ln |y| - \ln \left|1 - \frac{y}{K}\right| &= rt + c \end{aligned} \quad (5.55)$$

where c is an arbitrary constant of integration to be determined from the initial condition $y(0) = y_0$. We have already noted that if $0 < y_0 < K$, then y remains in this interval for all time. Thus in this case we can remove the absolute value bars in Eq. (4.55), and by taking the exponential of both sides, we find that

$$\frac{y}{1 - (y/K)} = Ce^{rt} \quad (5.56)$$

where $C = e^c$. In order to satisfy the initial condition $y(0) = y_0$, we must choose $C = \frac{y_0}{1 - (y_0/K)}$. Using this value of C in Eq. (4.56) and solving for y , we obtain

$$y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}} \quad (5.57)$$

We have derived the solution (4.57) under the assumption that $0 < y_0 < K$. If $y_0 > K$, then the details of dealing with Eq. (4.55) are only slightly different, and Eq. (4.57) is also valid in this case.

Note that Eq. (4.57) also contains the equilibrium solutions $y = \phi_1(t) = 0$ and $y = \phi_2(t) = K$ corresponding to the initial conditions $y_0 = 0$ and $y_0 = K$, respectively.

In particular, if $y_0 = 0$, then Eq. (4.57) requires that $y(t) = 0$ for all t . If $y_0 > 0$, and if we let $t \rightarrow \infty$ in Eq. (4.57), then we obtain

$$\lim_{t \rightarrow \infty} y(t) = \frac{y_0 K}{y_0} = K$$

Thus, for each $y_0 > 0$, the solution approaches the equilibrium solution $y = \phi_2(t) = K$ asymptotically as $t \rightarrow \infty$. Therefore we say that the solution $\phi_2(t) = K$ is an asymptotically stable solution of Eq. (4.53) or that the point $y = K$ is an asymptotically stable equilibrium or critical point.

After a long time, the population is close to the saturation level K regardless of the initial population size, as long as it is positive. Other solutions approach the equilibrium solution more rapidly as r increases.

On the other hand, the situation for the equilibrium solution $y = \phi_1(t) = 0$ is quite different. Even solutions that start very near zero grow as t increases and, as we have seen, approach K as $t \rightarrow \infty$. We say that $\phi_1(t) = 0$ is an unstable equilibrium solution or that $y = 0$ is an unstable equilibrium critical point. This means that the only way to guarantee that the solution remains near zero is to make sure its initial value is exactly equal to zero.

If we want to find the time, t , when the population reach certain level $y(t)$, e.g. when the tuna population in Pacific Ocean reach 85% ($y(t) = 0.85K$) of its maximum limit that the environment can carry, then we will use this formula from derivating of Eq. (4.57):

$$t = -\frac{1}{r} \ln \frac{\left(\frac{y_0}{K}\right) \left[1 - \left(\frac{y}{K}\right)\right]}{\left(\frac{y}{K}\right) \left[1 - \left(\frac{y_0}{K}\right)\right]} \quad (5.58)$$

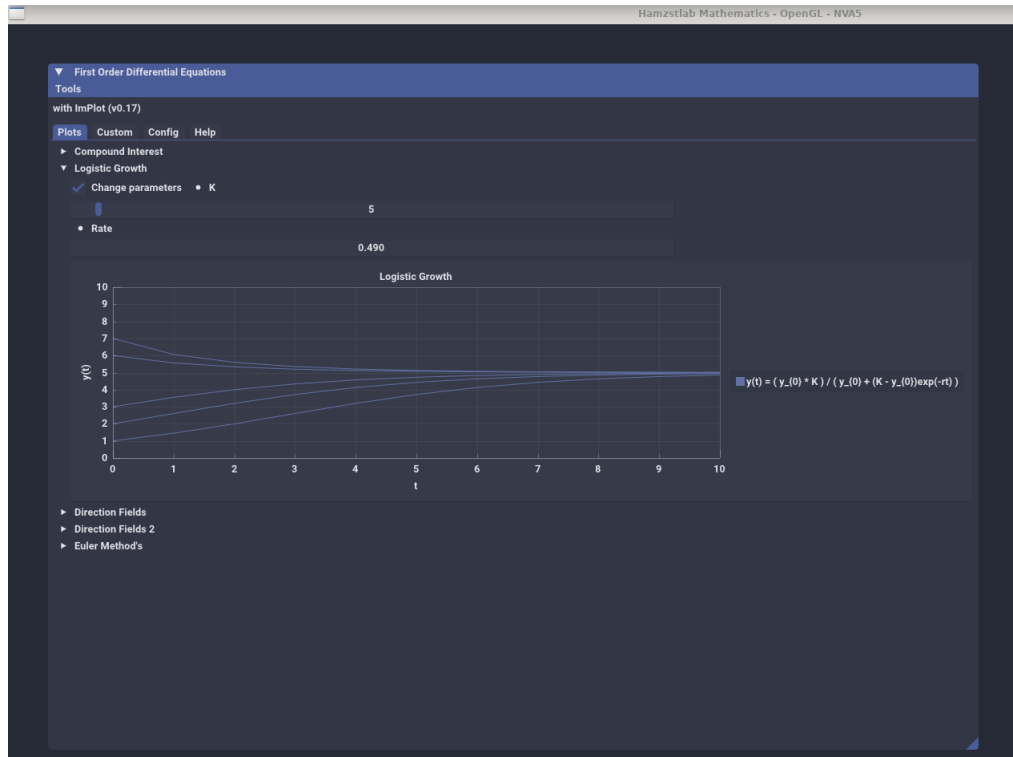


Figure 5.9: The solution curves for $y(t) = \frac{y_0 K}{y_0 + (K - y_0)e^{-rt}}$ we set y_0 with fixed values at first and the parameters K and r can be changed at in instant in Hamzstlab Mathematics ([Hamzstlab-Mathematics/examples/First Order Differential Equation/main.cpp](#)).

[H*] A Critical Threshold

We now turn to a consideration of the equation

$$\frac{dy}{dt} = -r \left(1 - \frac{y}{T} \right) y \quad (5.59)$$

where r and T are given positive constants. Observe that this equation differs from the logistic equation (4.53) in the presence of the minus sign on the right side, along with T replacing K .

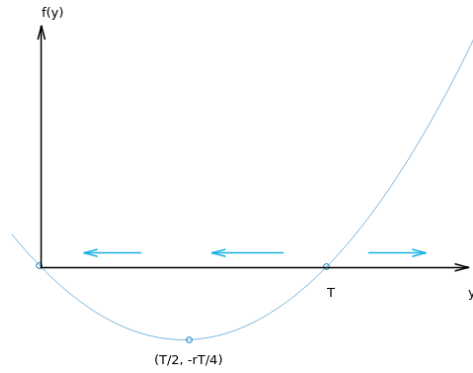


Figure 5.10: The graph of $f(y)$ for $\frac{dy}{dt} = -r \left(1 - \frac{y}{T}\right) y$ (Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Critical Threshold/main.cpp).

The intercepts on the y -axis are the critical points $y = 0$ and $y = T$, corresponding to the equilibrium solutions $\phi_1(t) = 0$ and $\phi_2(t) = T$.

The dots at $y = 0$ and $y = T$ are the critical points, or equilibrium solutions, and the arrows indicate where solutions are either increasing or decreasing.

If $0 < y < T$, then $\frac{dy}{dt} < 0$, and y decreases as T increases. Thus $\phi_1(t) = 0$ is an asymptotically stable equilibrium solution and $\phi_2(t) = T$ is an unstable one.

Further, $f'(y)$ is negative for $0 < y < T/2$ and positive for $T/2 < y < T$, so the graph of y versus t is concave up and concave down, respectively, in these intervals. Also, $f'(y)$ is positive for $y > T$, so the graph of y versus t is also concave up here.

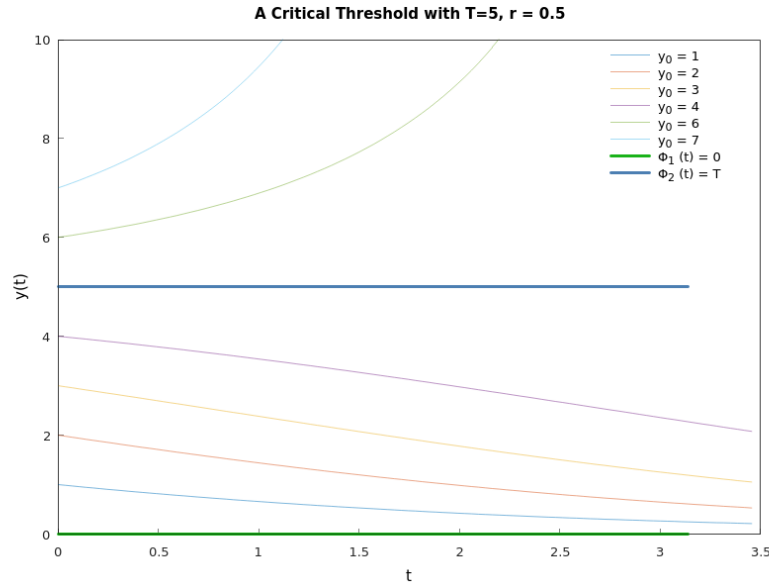


Figure 5.11: The solution curves for growth with a threshold for $\frac{dy}{dt} = -r \left(1 - \frac{y}{T}\right) y$ (Hamzstplot/Examples with Makefile/Plot 2D Solution Curves for Critical Threshold/main.cpp).

Figure (4.11) shows the solution curves of Eq. (4.59). After we draw the two thick horizontal line at $y = 0$ and $y = T$, we can then sketch the curves in the strip $0 < y < T$ that are decreasing as t increases and change concavity as they cross the line $y = T/2$.

Next draw some curves above $y = T$ that increase more and more steeply as t and y increase. Make sure that all curves become flatter as y approaches either zero or T .

From this figure it appears that as time increases, y either approaches zero or grows without bound, depending on whether the initial value y_0 is less than or greater than T . Thus T is a threshold level, below which growth does not occur.

By solving the differential equation (4.59) with separating the variables and integrating we will obtain

$$y(t) = \frac{y_0 T}{y_0 + (T - y_0)e^{rt}} \quad (5.60)$$

which is the solution of Eq. (4.59) subject to the initial condition $y(0) = y_0$.

If $0 < y_0 < T$, then it follows from Eq. (4.60) that $y \rightarrow 0$ as $t \rightarrow \infty$. This agrees with our qualitative geometric analysis. If $y_0 > T$, then the denominator on the right side of Eq. (4.60) is zero for a certain finite value of t . We denote this value by t^* and calculate it from

$$y_0 - (y_0 - T)e^{rt^*} = 0$$

which gives

$$t^* = \frac{1}{r} \ln \frac{y_0}{y_0 - T} \quad (5.61)$$

Thus, if the initial population y_0 is above the threshold T , the threshold model predicts that the graph of y versus t has a vertical asymptote at $t = t^*$; in other words, the population becomes unbounded in a finite time, whose value depends on y_0 , T , and r . The existence and location of this asymptote were not apparent from the geometric analysis, so in this case the explicit solution yields additional important qualitative, as well as quantitative, information.

The population of some species exhibit the threshold phenomenon. If too few are present, then the species cannot propagate itself successfully and the population becomes extinct. However, if the population is alrger than the threshold level, then further growth occurs. Of course, the population cannot become unbounded, so eventually Eq. (4.59) must be modified to take this into account.

Critical thresholds also occur in other circumstances. For example, in fluid mechanics, equations of the form (4.53) and (4.59) often govern the evolution of a small disturbance y in a laminar (or smooth) fluid flow.

For instance, if Eq. (4.59) holds and $y < T$, then the disturbance is damped out and laminar flow persists.

However, if $y > T$, then the disturbance grows larger and the laminar flow breaks up into a turbulent one. In this case T is referred to as the critical amplitude. Experimenters speak of keeping the disturbance level in a wind tunnel sufficiently low so taht they can study laminar flow over an airfoil, for example.

[H*] Logistic Growth with a Threshold

This model comes after we modified the threshold model (4.59) so that unbounded growth does not occur when y is above the threshold T . The simplest way to do this is to introduce another factor that will have the effect of making $\frac{dy}{dt}$ negative when y is large. Thus we consider

$$\frac{dy}{dt} = -r \left(1 - \frac{y}{T}\right) \left(1 - \frac{y}{K}\right) y \quad (5.62)$$

where $r > 0$ and $0 < T < K$.

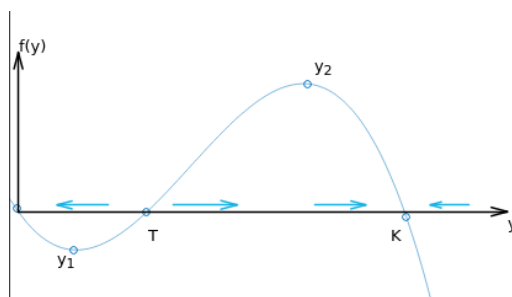


Figure 5.12: The graph of $f(y)$ versus y for logistic growth with a threshold for $\frac{dy}{dt} = -r \left(1 - \frac{y}{T}\right) \left(1 - \frac{y}{K}\right) y$ (Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Logistic Growth with a Threshold/main.cpp).

The equilibrium solutions are found by setting $\frac{dy}{dt} = 0$.

In this problem there are three critical points, $y = 0$, $y = T$, and $y = K$, corresponding to the equilibrium solutions $\phi_1(t) = 0$, $\phi_2(t) = T$, and $\phi_3(t) = K$, respectively.

From Figure 4.12 we observe that $\frac{dy}{dt} > 0$ for $T < y < K$, and consequently y is increasing there. Further, $\frac{dy}{dt} < 0$ for $y < T$ and for $y > K$, so y is decreasing in these intervals.

Consequently, the equilibrium solutions $\phi_1(t)$ and $\phi_3(t)$ are asymptotically stable, and the solution $\phi_2(t)$ is unstable.

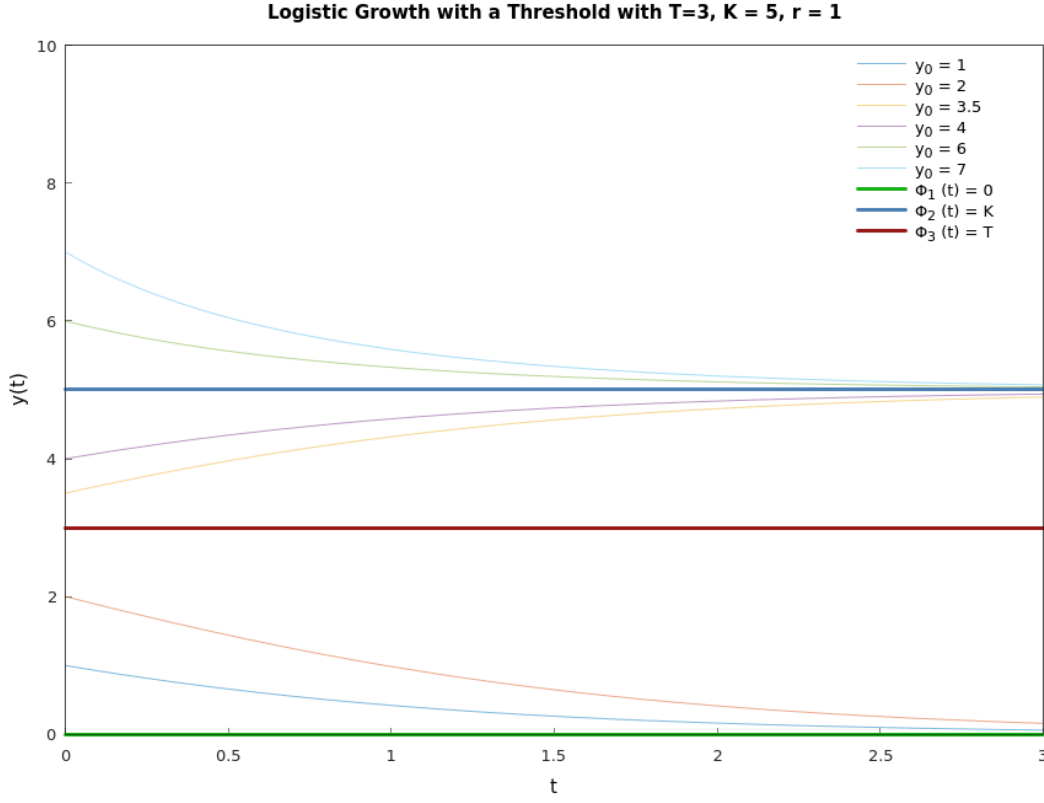


Figure 5.13: The solution curves for logistic growth with a threshold for $\frac{dy}{dt} = -r \left(1 - \frac{y}{T}\right) \left(1 - \frac{y}{K}\right) y$ (Hamzstplot/Examples with Makefile/Plot 2D Solution Curves for Logistic Growth with a Threshold/main.cpp).

You should make sure that you understand the relation between these two Figure (4.12) and Figure (4.13).

We see that if y starts below the threshold T , then y declines to ultimate extinction. On the other hand, if y starts above T , then y eventually approaches the carrying capacity K . The inflection points on the graphs of y versus t in Figure (4.13) correspond to the maximum and minimum points y_1 and y_2 , respectively, on the graph of $f(y)$ versus y in Figure (4.12). These values can be obtained by differentiating the right side of Eq. (4.62) with respect to y , setting the result equal to zero, and solving for y . We obtain

$$y_{1,2} = \frac{\left(K + T \pm \sqrt{K^2 - KT + T^2}\right)}{3} \quad (5.63)$$

where the plus sign yields y_1 and the minus sign yields y_2 .

The logistic equation with a threshold population at Eq. (4.62) can be solved using the method of separation of variables. However, it is very difficult to get the solution as an explicit function of t , so we are opting for the implicit function instead.

$$\begin{aligned}\frac{dy}{dt} &= -r \left(1 - \frac{y}{T}\right) \left(1 - \frac{y}{K}\right) y \\ \frac{dy}{\left(1 - \frac{y}{T}\right) \left(1 - \frac{y}{K}\right) y} &= -r dt \\ \ln |y| - \ln |T - y| - \ln |K - y| &= -rt + C\end{aligned}$$

Solving for C with the initial value of $y(0) = y_0$, we will obtain

$$C = \ln |y_0| - \ln |T - y_0| - \ln |K - y_0|$$

thus

$$\ln |y(t)| - \ln |T - y(t)| - \ln |K - y(t)| = -rt + \ln |y_0| - \ln |T - y_0| - \ln |K - y_0| \quad (5.64)$$

where

- * $y(t)$ represents the population size at time t .
- * r is a positive constant representing the intrinsic growth rate.
- * T is the threshold population below which the population declines towards extinction.
- * K is the carrying capacity, the maximum sustainable population size.
- * $0 < T < K$

The good thing is, sometimes you can solve a very complex problem by dividing it into smaller partition and see the pattern at each partition, in C++ code we can use **if.. then** conditional that can be very useful for drawing the solution curves of this type. Without even solving the Eq. (4.62), we can use Eq. (4.57) and Eq. (4.60) under the **if .. then .. if else .. then ..** conditional.

[H*] Phase Line Analysis

In order to determine the stability for single variable function, we use phase line analysis.

First, find the equilibrium solutions by finding the root of $f(y) = 0$ from

$$\frac{dy}{dt} = f(y)$$

Then, draw a horizontal line representing the variable (e.g. y). Mark the equilibrium points on the line.

Analyze the sign of the derivative $\frac{dy}{dt}$ on either side of each equilibrium point.

If the derivative changes from positive to negative as y increases, the equilibrium point is a sink (stable).

If it changes from negatives to positive, it's a source (unstable).

If there's no sign change, it's a node (semistable).

To draw the phase line you can just rotate the horizontal line from above counterclockwise 90° to become a vertical line.

The benefits of phase line analysis:

1. Phase lines provide a visual understanding of the system's behavior without solving the differential equation.
2. They clearly show which equilibrium points are stable (solutions approach them) and which are unstable (solutions move away).
3. They reveal how solutions behave as time approaches infinity.

[H*] Harvesting a Renewable Resource

Suppose that the population of y of a certain species of fish (for example, tuna or halibut) in a given area of the ocean is described by the logistic equation

$$\frac{dy}{dt} = r \left(1 - \frac{y}{K} \right) y \quad (5.65)$$

Although it is desirable to utilize this source of food, it is intuitively clear that if too many fish are caught, then the fish population may be reduced below a useful level and possibly even driven to extinction.

The solution to Eq. (4.65) with initial value problem $y(0) = y_0$ is

$$y(t) = \frac{y_0 K e^{rt}}{(y_0 - K) \left(\frac{y_0 e^{rt}}{y_0 - K} - 1 \right)} \quad (5.66)$$

[H*] Example:

At a given level of effort, it is reasonable to assume that the rate at which fish are caught depends on the population y : the more fish there are, the easier it is to catch them. Thus we assume that the rate at which fish are caught is given by Ey , where E is a positive constant, with units of 1/time, that measures the total effort made to harvest the given species of fish. To include this effect, the logistic equation is replaced by

$$\frac{dy}{dt} = r \left(1 - \frac{y}{K} \right) y - Ey \quad (5.67)$$

This equation is known as the Schaefer model after the biologist M.B. Schaefer, who applied it to fish populations.

- (a) Show that if $E < r$, then there are two equilibrium points $y_1 = 0$ and $y_2 = K \left(1 - \frac{E}{r} \right) > 0$.
- (b) Determine the stability for the equilibrium points.
- (c) A sustainable yield Y of the fishery is a rate at which fish can be caught indefinitely. It is the product of the effort E and the asymptotically stable population y_2 . Find Y as a function of the effort E ; the graph of this function is known as the yield-effort curve.
- (d) Determine E so as to maximize Y and thereby find the maximum sustainable yield Y_m .

Solution:

- (a) To determine the equilibrium points or the critical points we need to locate the roots of $f(y) = 0$

$$\begin{aligned} f(y) &= 0 \\ r \left(1 - \frac{y}{K} \right) y - Ey &= 0 \\ -\frac{ry^2}{K} + (r - E)y &= 0 \end{aligned}$$

the roots are

$$\begin{aligned}
 y_{1,2} &= \frac{-(r-E) \pm \sqrt{(r-E)^2 - 4\left(-\frac{r}{K}\right)(0)}}{2\left(-\frac{r}{K}\right)} \\
 &= \frac{-(r-E) \pm (r-E)}{-\frac{2r}{K}} \\
 y_{1,2} &= \begin{cases} 0 \\ \frac{-2r+2E}{-2r/K} = \frac{K(E-r)}{-r} = K\left(1 - \frac{E}{r}\right) \end{cases}
 \end{aligned}$$

If $E < r$ then $K\left(1 - \frac{E}{r}\right) > 0$ and is a possible equilibrium population.

- (b) To determine which equilibrium solution that is stable and unstable we will check with inputting y as several values in $\frac{dy}{dt} = f(y)$

$$\begin{aligned}
 f(y) &= -\frac{ry^2}{K} + (r-E)y \\
 &= \left(y - K\left(1 - \frac{E}{r}\right)\right)y
 \end{aligned}$$

If $0 < y < K\left(1 - \frac{E}{r}\right)$ then $\frac{dy}{dt} < 0$, so y is decreasing on these intervals.

If $y > K\left(1 - \frac{E}{r}\right)$ then $\frac{dy}{dt} > 0$, so y is increasing on these intervals.

The equilibrium at $y = 0$ is a sink (asymptotically stable) and the equilibrium at $K\left(1 - \frac{E}{r}\right)$ is a source (unstable).

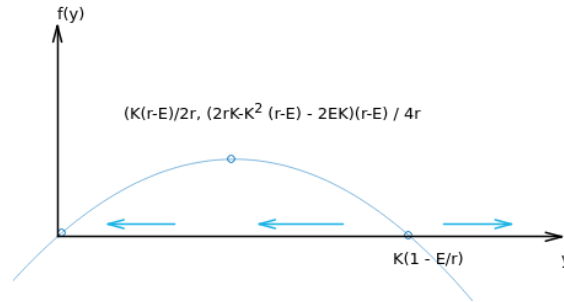


Figure 5.14: The equilibrium points and the stability analysis for $\frac{dy}{dt} = f(y) = \left(y - K\left(1 - \frac{E}{r}\right)\right)y$ (Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Schaefer Model/main.cpp).

To determine the vertex of the parabola you will have to compute

$$\begin{aligned}
 \frac{df(y)}{dy} &= 0 \\
 \frac{d\left(-\frac{ry^2}{K} + (r-E)y\right)}{dy} &= 0 \\
 y &= \frac{K(r-E)}{2r}
 \end{aligned}$$

then input y back into $f(y)$ again to obtain the vertex fully

$$\begin{aligned} f(y) &= -\frac{r \left(\frac{K(r-E)}{2r} \right)^2}{K} + (r-E) \left(\frac{K(r-E)}{2r} \right) \\ &= \frac{(2rK - K^2(r-E) - 2EK)(r-E)}{4r} \end{aligned}$$

thus the vertex is $\left(\frac{K(r-E)}{2r}, \frac{(2rK - K^2(r-E) - 2EK)(r-E)}{4r} \right)$.

(c) The sustainable yield is the effort times the asymptotically stable population.

$$Y(E) = EK \left(1 - \frac{E}{r} \right)$$

To maximize the sustainable yield, take the derivative of it, set it equal to zero, and solve the resulting equation for E .

$$\begin{aligned} Y'(E) &= 0 \\ K \left(1 - \frac{E}{r} \right) + EK \left(-\frac{1}{r} \right) &= 0 \\ K \left(1 - \frac{E}{r} \right) &= \frac{EK}{r} \\ K(r-E) &= EK \\ rK - EK &= EK \\ 2EK &= rK \\ E &= \frac{r}{2} \end{aligned}$$

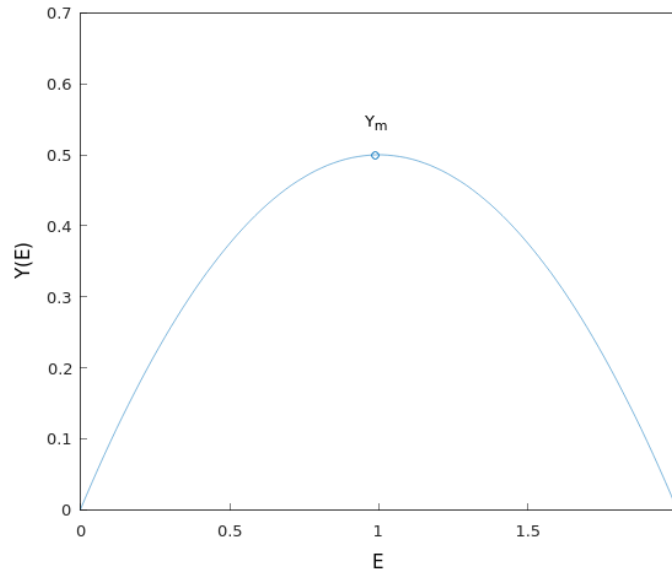


Figure 5.15: The yield-effort curve for $Y(E) = EK \left(1 - \frac{E}{r} \right)$ with $r = 2$ and $K = 1$ (Hamzstplot/Examples with Makefile/Plot 2D Yield-Effort Curve for Schaefer Modell/main.cpp).

(d) The maximum sustainable yield is therefore

$$\begin{aligned} Y_m &= Y\left(E = \frac{r}{2}\right) \\ &= \left(\frac{r}{2}\right) K \left(1 - \frac{1}{2}\right) \\ &= \frac{rK}{4} \end{aligned}$$

[H*] Example:

We assume that fish are caught at a constant rate h independent of the size of the fish population. Then y satisfies

$$\frac{dy}{dt} = r \left(1 - \frac{y}{K}\right) y - h \quad (5.68)$$

The assumption of a constant catch rate h may be reasonable when y is large but becomes less so when y is small.

- If $h < rK/4$, show that Eq. (4.68) has two equilibrium points y_1 and y_2 with $y_1 < y_2$; determine these points.
- Show that y_1 is unstable and y_2 is asymptotically stable.
- From a plot of $f(y)$ versus y , show that if the initial population $y_0 > y_1$, then $y \rightarrow y_2$ as $t \rightarrow \infty$, but that if $y_0 < y_1$, then y decreases as t increases. Note that $y = 0$ is not an equilibrium point, so if $y_0 < y_1$, then extinction will be reached in a finite time.
- If $h > rK/4$, show that y decreases to zero as t increases regardless of the value of y_0 .
- If $h = rK/4$, show that there is a single equilibrium point $y = K/2$ and that this point is semistable. Thus the maximum sustainable yield is $h_m = rK/4$, corresponding to the equilibrium value $y = K/2$. Observe that h_m has the same value as Y_m in the previous example. The fishery is considered to be overexploited if y is reduced to a level below $K/2$.

Solution:

- To determine the equilibrium points we will find the roots of $f(y) = 0$

$$\begin{aligned} f(y) &= 0 \\ r \left(1 - \frac{y}{K}\right) y - h &= 0 \\ -\frac{r}{K} y^2 + ry - h &= 0 \end{aligned}$$

with quadratic roots we will obtain

$$\begin{aligned} y_{1,2} &= \frac{-r \pm \sqrt{r^2 - 4 \left(-\frac{r}{K}\right) (-h)}}{2 \left(-\frac{r}{K}\right)} \\ &= \frac{-r \pm \sqrt{r^2 - \left(\frac{4rh}{K}\right)}}{-\frac{2r}{K}} \\ &= \frac{r \pm \sqrt{r^2 - \left(\frac{4rh}{K}\right)}}{\frac{2r}{K}} \end{aligned}$$

$$y_{1,2} = \begin{cases} \frac{r + \sqrt{r^2 - \left(\frac{4rh}{K}\right)}}{\frac{2r}{K}} = \frac{K + \sqrt{K^2 - \frac{4Kh}{r}}}{2} \\ \frac{r - \sqrt{r^2 - \left(\frac{4rh}{K}\right)}}{\frac{2r}{K}} = \frac{K - \sqrt{K^2 - \frac{4Kh}{r}}}{2} \end{cases}$$

These are the equilibrium solutions as long as

$$\begin{aligned} r^2 - \frac{4rh}{K} &> 0 \\ h &< \frac{rK}{4} \end{aligned}$$

(b) We will try to plot $f(y) = r \left(1 - \frac{y}{K}\right) y - h$ versus y with $r = 1, K = 1$, and $h = 1/8$.

With the chosen parameters, we will have

$$f(y) = (1 - y)y - \frac{1}{8}$$

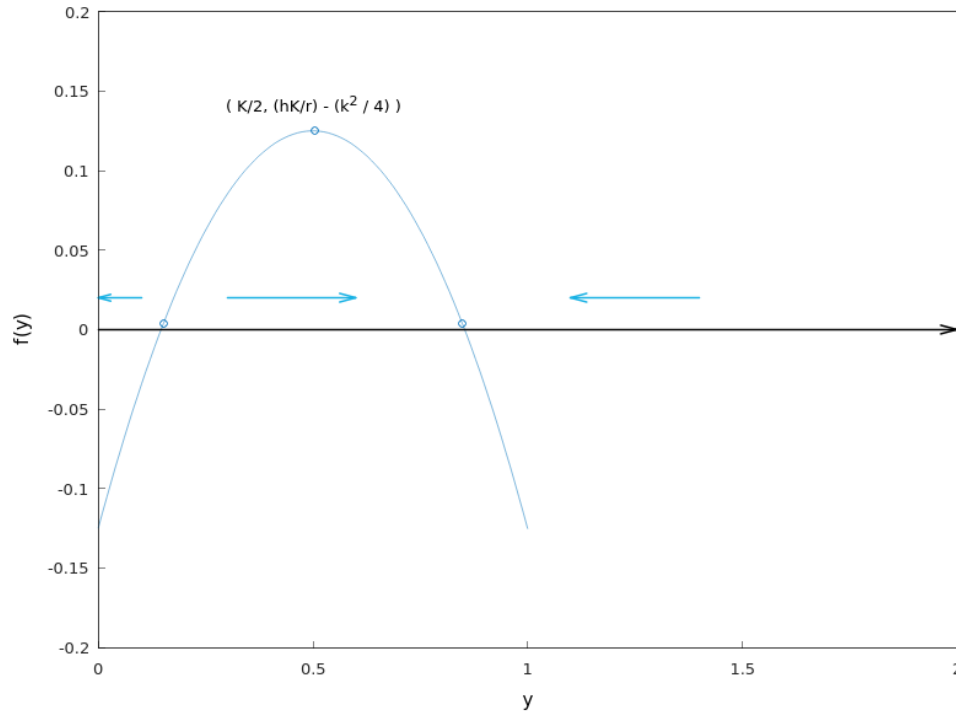


Figure 5.16: The plot for $f(y) = (1 - y)y - \frac{1}{8}$ (Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Schaefer Model Variation/main.cpp).

To save more time, we do not need to compute the exact numerical equilibrium points of y_1 and y_2 here, we only need to know $f(y) = (1 - y)y - \frac{1}{8}$ and then plot $f(y)$ versus y , from the plot we will be able to determine the stability by approximation. For example, it looks very clearly that $y = 0$ is less than $y = y_1$ so we can compute $f(0)$ to see $\frac{dy}{dt}$ behavior in that interval, if $y < y_1$ then $\frac{dy}{dt} < 0$ and the arrow

is pointing to the left on the horizontal line, or pointing downward in the phase line.

We see that the equilibrium point y_1 is a source (unstable) and the equilibrium point at $y = y_2$ is a sink (asymptotically stable).

(c) Suppose that the initial condition is

$$y(0) = y_0$$

As indicated by the arrows from Figure (4.16), if $y_0 > y_1$, then the population will tend to $y(t) = y_2$ as $t \rightarrow \infty$.

If $y_0 = y_1$, then the population will remain there.

If $y_0 < y_1$, then the population will tend to zero in a finite time because there are no equilibrium points to the left of $y = y_1$.

(d) Suppose that $h > \frac{rK}{4}$, for example, using $r = 1$, $K = 1$, and $h = 1$. Then the entire graph of $f(y) = r \left(1 - \frac{y}{K}\right) y - h$ versus y lies below the y -axis, which means all the arrows point to the left.

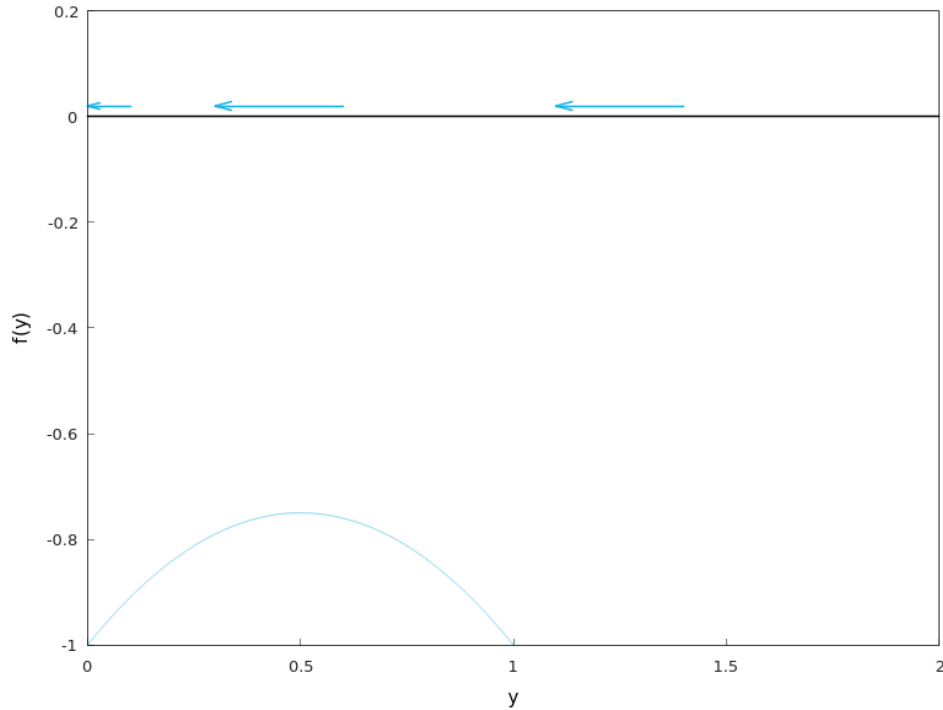


Figure 5.17: The plot for $f(y) = (1 - y)y - 1$ (Hamzstplot/Examples with Makefile/Plot 2D No Equilibrium Solution for Schaefer Model Variation/main.cpp).

This happens because $h > \frac{rK}{4}$ and this condition does not fit the requirement to have equilibrium solution from part (a).

The population tends to zero regardless of what y_0 is.

(e) Suppose that $h = \frac{rK}{4}$, then there will only be one equilibrium point

$$y = \frac{K \pm \sqrt{K^2 - \frac{4Kh}{r}}}{2} = \frac{K \pm \sqrt{0}}{2} = \frac{K}{2}$$

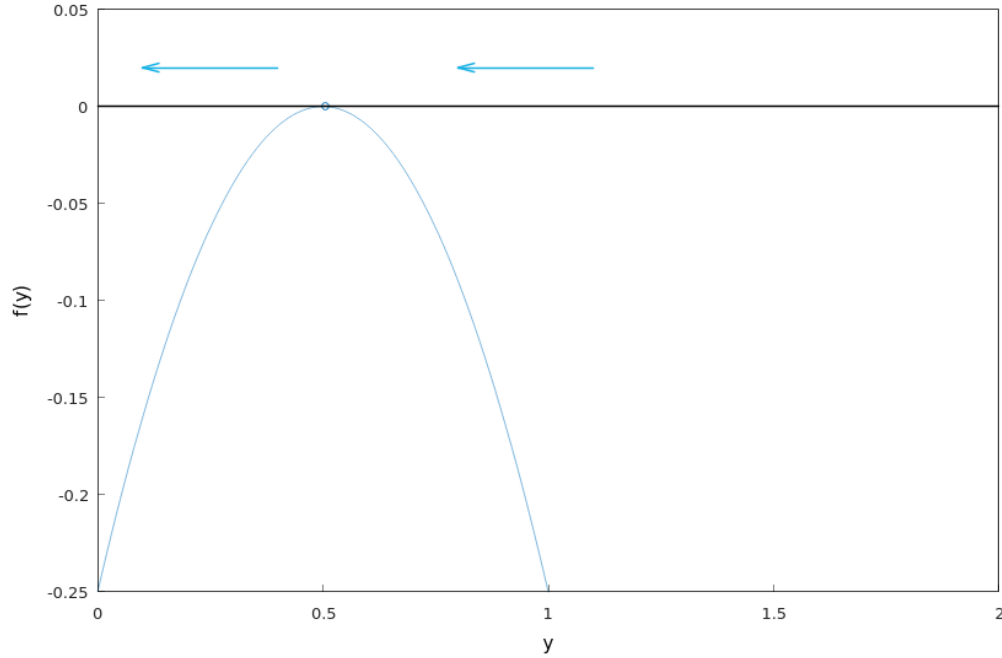


Figure 5.18: The plot for $f(y) = (1 - y)y - \frac{1}{4}$ (Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution Only 1 for Schaefer Model Variation/main.cpp).

Even though the graph lies below the y -axis, the difference between part (d) is that there's now an equilibrium point at $y = \frac{K}{2}$.

If $y_0 > \frac{K}{2}$, then the population will tend to $y(t) = \frac{K}{2}$ as $t \rightarrow \infty$.

If $y_0 = \frac{K}{2}$, then the population will remain there.

If $y_0 < \frac{K}{2}$, then the population will tend to zero in a finite time because there are no equilibria to the left of $y = \frac{K}{2}$.

Thus, the equilibrium point here at $y = \frac{K}{2}$ is semistable (stable from one side and unstable from the other side).

[H*] Epidemics

The use of mathematical methods to study the spread of contagious diseases goes back at least to some work by Daniel Bernoulli in 1760 on smallpox. In more recent years many mathematical models have been proposed and studied for many different diseases. Similar models have also been used to describe the spread of rumors and of

consumer products.

[H*] Example:

Suppose that a given population can be divided into two parts: those who have a given disease and can infect others, and those who do not have it but are susceptible. Let x be the proportion of susceptible individuals and y the proportion of infectious individuals; then $x + y = 1$. Assume that the disease spreads by contact between sick and well members of the population and that the rate of spread $\frac{dy}{dt}$ is proportional to the number of such contacts. Further, assume that members of both groups move about freely among each other, so the number of contacts is proportional to the product of x and y . Since $x = 1 - y$, we obtain the initial value problem

$$\frac{dy}{dt} = \alpha y(1 - y), \quad y(0) = y_0 \quad (5.69)$$

where α is positive proportionality factor, and y_0 is the initial proportion of infectious individuals.

- (a) Find the equilibrium points for the differential equation (4.64) and determine whether each is asymptotically stable, semistable, or unstable.
- (b) Solve the initial value problem (4.69) and verify that the conclusions you reached in part (a) are correct. Show that $y(t) \rightarrow 1$ as $t \rightarrow \infty$, which means that ultimately the disease spreads through the entire population.

Solution:

- (a) The equilibrium points are found by setting

$$\frac{dy}{dt} = 0$$

and solving the resulting equation for y .

$$\alpha y(1 - y)$$

The equilibrium points are at $y = \phi_1(t) = 0$ and $y = \phi_2(t) = 1$. With phase line analysis we will obtain that the equilibrium solution $y = \phi_1(t) = 0$ is unstable and the equilibrium solution $y = \phi_2(t) = 1$ is asymptotically stable.

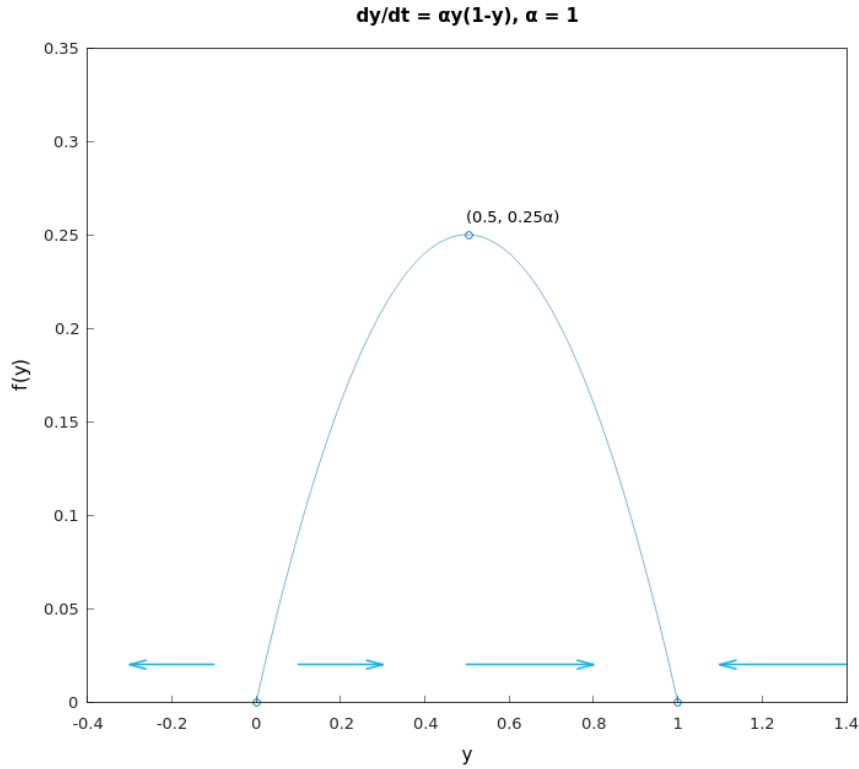


Figure 5.19: The plot for $f(y) = \alpha y(1 - y)$ with $\alpha = 1$ (Hamzstplot/Examples with Makefile/Plot 2D Equilibrium Solution for Smallpox Epidemics Modell/main.cpp).

What this means is that even if only one individual is infected in a population of eight billion, the entire population will eventually be infected.

(b) The ODE is separable, with

$$[y(1 - y)]^{-1} dy = \alpha dt$$

Integrating both sides and invoking the initial condition, the solution is

$$\begin{aligned} \frac{dy}{\alpha y(1 - y)} &= dt \\ \frac{1}{\alpha y} + \frac{1/\alpha}{1 - y} dy &= dt \\ \frac{1}{\alpha} \ln |y| + \frac{1}{\alpha y} (-\ln |y - 1|) &= t + c \\ \frac{\ln |y| - \ln |y - 1|}{\alpha} &= t + c \\ \ln \left| \frac{y}{y - 1} \right| &= \alpha t + C \end{aligned}$$

To determine C we will apply the initial condition with $y(0) = y_0$, thus

$$C = \ln \left| \frac{y_0}{y_0 - 1} \right|$$

As a result, the previous equation becomes

$$\ln \left| \frac{y}{y-1} \right| = \alpha t + \ln \left| \frac{y_0}{y_0-1} \right|$$

Since y and y_0 are between 0 and 1, the absolute value signs can be removed by writing the denominators as $1 - y$ and $1 - y_0$.

$$\begin{aligned} \ln \frac{y}{1-y} &= \alpha t + \ln \frac{y_0}{1-y_0} \\ \frac{y}{1-y} &= e^{\alpha t + \ln \frac{y_0}{1-y_0}} \\ &= e^{\alpha t} e^{\ln \frac{y_0}{1-y_0}} \\ &= e^{\alpha t} \left(\frac{y_0}{1-y_0} \right) \\ \frac{y}{1-y} (1-y) &= (1-y) \left(e^{\alpha t} \left(\frac{y_0}{1-y_0} \right) \right) \\ y &= e^{\alpha t} \left(\frac{y_0}{1-y_0} \right) - e^{\alpha t} \left(\frac{y_0}{1-y_0} \right) y \\ y \left[1 + e^{\alpha t} \left(\frac{y_0}{1-y_0} \right) \right] &= e^{\alpha t} \left(\frac{y_0}{1-y_0} \right) \\ y &= \frac{e^{\alpha t} \left(\frac{y_0}{1-y_0} \right)}{1 + e^{\alpha t} \left(\frac{y_0}{1-y_0} \right)} \\ &= \frac{e^{\alpha t} y_0}{(1-y_0) + e^{\alpha t} y_0} \\ y(t) &= \frac{y_0}{(1-y_0)e^{-\alpha t} + y_0} \end{aligned}$$

It is evident that (independent of y_0)

$$\lim_{t \rightarrow -\infty} y(t) = 0$$

and

$$\lim_{t \rightarrow \infty} y(t) = 1$$

If there are no infected individuals initially ($y_0 = 0$), then $y(t) = 0$. However, if $0 < y_0 < 1$, then the exponential term decays to zero as $t \rightarrow \infty$. If at least one person is infected, then ultimately the disease spreads through the entire population.

[H*] Bifurcation Points

For an equation of the form

$$\frac{dy}{dt} = f(a, y) \tag{5.70}$$

where a is a real parameter, the critical points (equilibrium solutions) usually depend on the value of a . As a steadily increases or decreases, it often happens that at a certain value of a , called a bifurcation point, critical points come together, or separate, and

equilibrium solutions may either be lost or gained.

Bifurcation points are of great interest in many applications, because near them the nature of the solution of the underlying differential equation is undergoing an abrupt change. For example, in fluid mechanics a smooth (laminar) flow may break up and become turbulent. Or an axially loaded column may suddenly buckle and exhibit a large lateral displacement. Or, as the amount of one of the chemicals in a certain mixture is increased, a spiral wave patterns of varying color may suddenly emerge in an originally quiescent fluid.

[H*] Example:

Consider the equation

$$\frac{dy}{dt} = a - y^2 \quad (5.71)$$

- (a) Find all of the critical points for Eq. (4.71). Observe that there are no critical points if $a < 0$, one critical point if $a = 0$, and two critical points if $a > 0$
- (b) Draw the phase line in each case and determine whether each critical point is asymptotically stable, semistable, or unstable.
- (c) In each case sketch several solutions of Eq. (4.71) in the ty -plane.
- (d) If we plot the location of the critical points as a function of a in the ay -plane, we obtain the bifurcation diagram for Eq. (4.71). The bifurcation at $a = 0$ is called a saddle-node bifurcation.

Solution:

- (a) The critical points are found by setting $\frac{dy}{dt} = 0$ and solving the resulting equation for y .

$$\begin{aligned} a - y^2 &= 0 \\ y^2 &= a \\ y &= \pm\sqrt{a} \end{aligned}$$

If a is negative, then there are no critical points; if a is zero, then there is one critical point; and if a is positive, then there are two critical points.

- (b) The phase line for the case of $a < 0$:

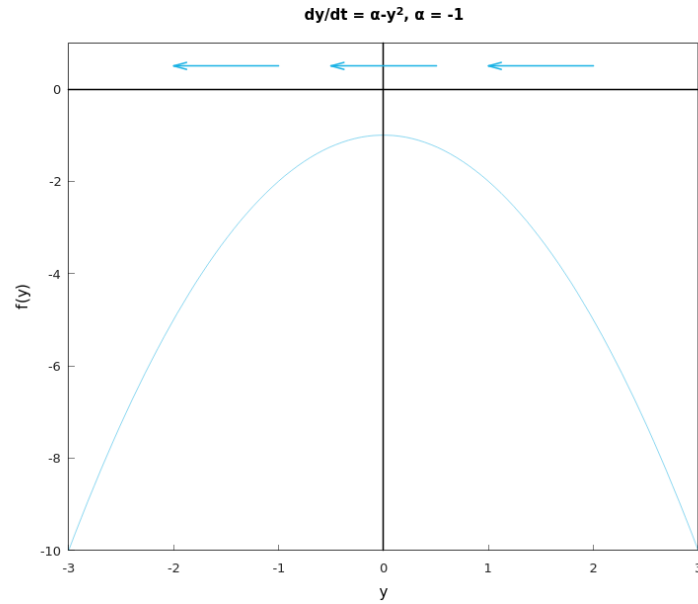


Figure 5.20: The plot for $f(y) = \alpha - y^2$ with $\alpha = -1$, the phase line is drawn with the blue arrows (*Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 25 No Equilibrium Solution/main.cpp*).

There is no critical point here.

The phase line for the case of $\alpha = 0$:

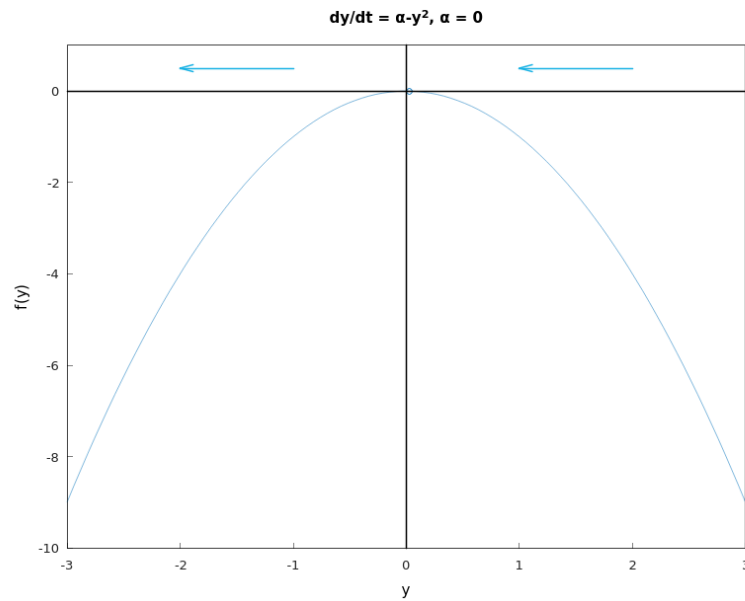


Figure 5.21: The plot for $f(y) = \alpha - y^2$ with $\alpha = 0$, the phase line is drawn with the blue arrows (*Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 25 Only 1 Equilibrium Solution/main.cpp*).

There is one critical point here $y = \phi_1(t) = 0$, and it is semistable.

The phase line for the case of $a > 0$:

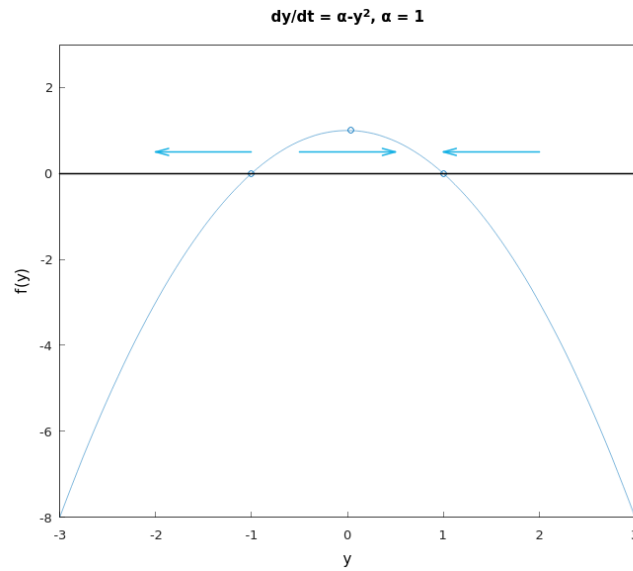


Figure 5.22: The plot for $f(y) = \alpha - y^2$ with $\alpha = 1$, the phase line is drawn with the blue arrows (*Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 25 2 Equilibrium Solutions/main.cpp*).

There are two critical points here $y = \phi_1(t) = \sqrt{a} = 1$ that is asymptotically stable and $y = \phi_2(t) = -\sqrt{a} = -1$ that is unstable.

(c) The ty -plane is the direction field for $\frac{dy}{dt}$

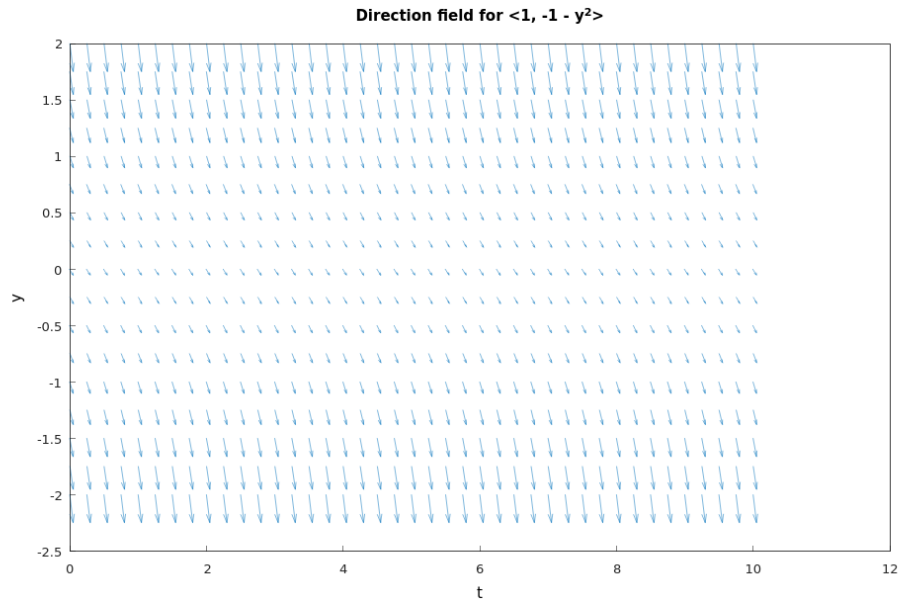


Figure 5.23: The direction field for $\frac{dy}{dt} = -1 - y^2$. Solution curves in the ty -plane corresponding to $\alpha = -1$ are tangent to the vectors in the direction field $\langle 1, -1 - y^2 \rangle$ at every point (*Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25a/main.cpp*).

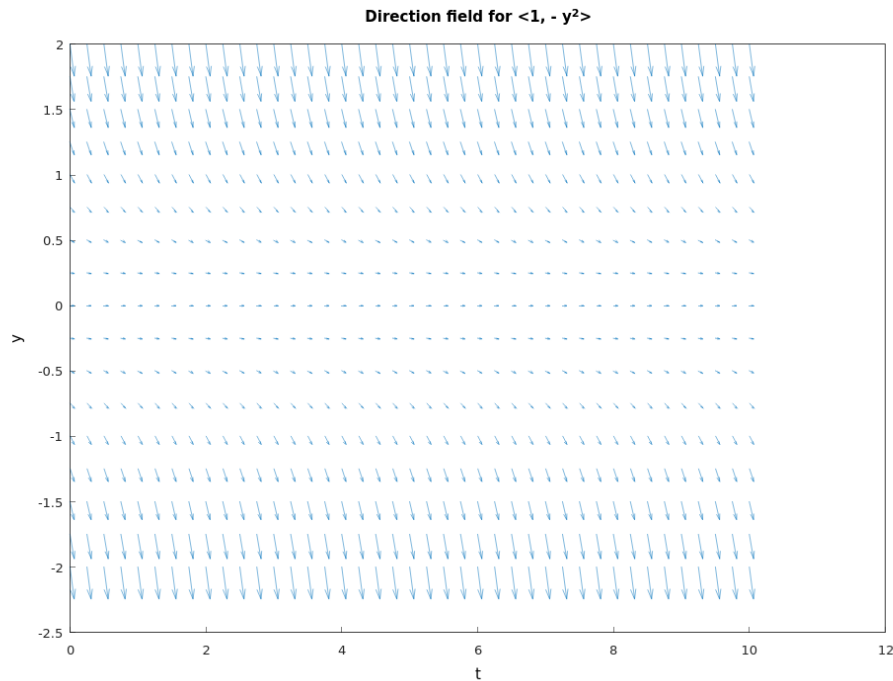


Figure 5.24: The direction field for $\frac{dy}{dt} = -y^2$. Solution curves in the ty -plane corresponding to $\alpha = 0$ are tangent to the vectors in the direction field $\langle 1, -y^2 \rangle$ at every point (*Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25b/main.cpp*).

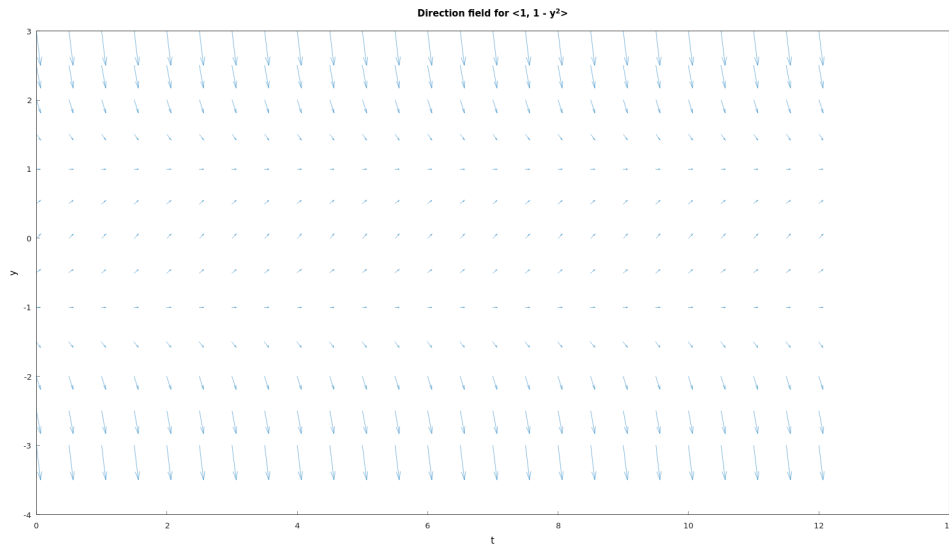


Figure 5.25: The direction field for $\frac{dy}{dt} = 1 - y^2$. Solution curves in the ty -plane corresponding to $\alpha = 1$ are tangent to the vectors in the direction field $\langle 1, 1 - y^2 \rangle$ at every point (Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp).

(d) We are going to plot the location of the critical points as a function of a , it is

$$y(a) = \pm\sqrt{a}$$

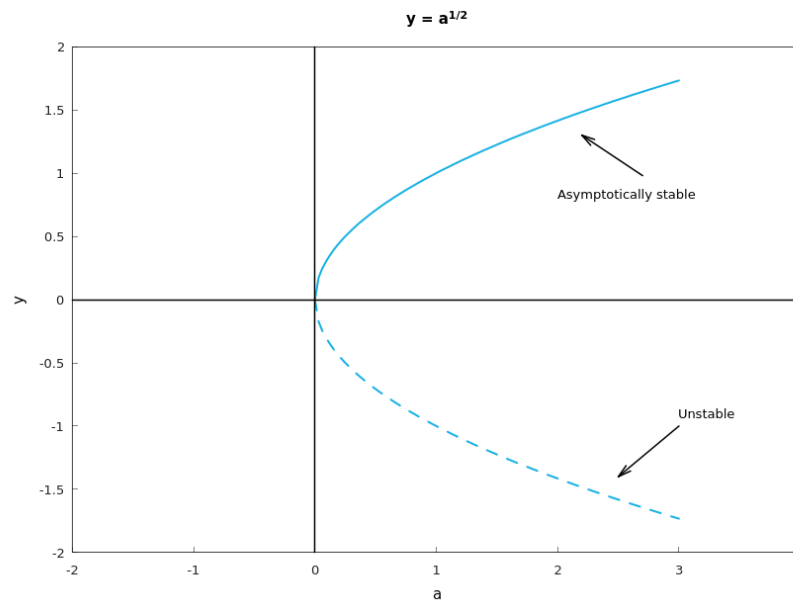


Figure 5.26: The bifurcation diagram for $\frac{dy}{dt} = a - y^2$, the bifurcation point at $a = 0$ is called a saddle-node bifurcation where the solution then differ into two kind of solution: stable and unstable (Hamzstplot/Examples with Makefile/Plot 2D Bifurcation Diagram Problem 25/main.cpp).

[H*] Example:

Consider the equation

$$\frac{dy}{dt} = ay - y^3 = y(a - y^2) \quad (5.72)$$

- Consider the cases $a < 0$, $a = 0$, and $a > 0$. In each case find all of the critical points for Eq. (4.72), draw the phase line, and determine whether each critical point is asymptotically stable, semistable, or unstable.
- In each case sketch several solutions of Eq. (4.72) in the ty -plane.
- Draw the bifurcation diagram of Eq. (4.72), that is, plot the location of the critical points versus a . For Eq. (4.72) the bifurcation point at $a = 0$ is called a pitchfork bifurcation.

Solution:

- The critical points are found by setting $\frac{dy}{dt} = 0$ and solving the resulting equation for y .

$$y(a - y^2) = 0$$

$$y = \{0, \pm\sqrt{a}\}$$

If a is negative, then there is only one critical point at $y = 0$; if a is zero, then there is still only one critical point at $y = 0$; and if a is positive, then there are three critical points.

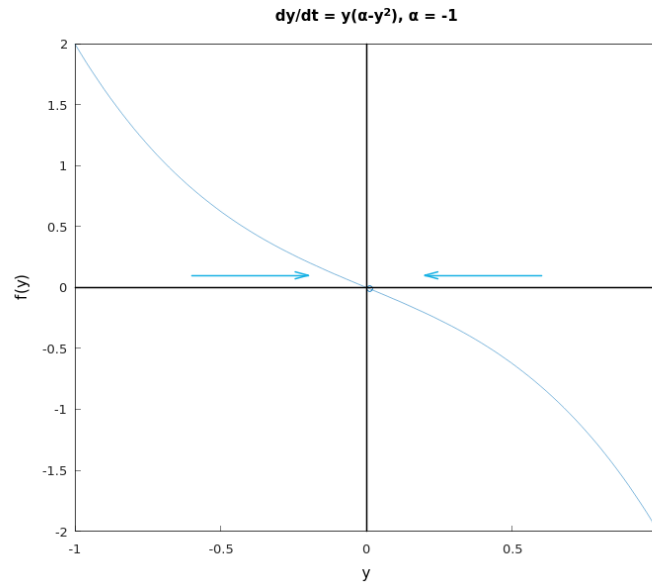


Figure 5.27: The plot for $f(y) = y(\alpha - y^2)$ with $\alpha = -1$, the phase line is drawn with the blue arrows (*Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 26 Only 1 Equilibrium Solution/main.cpp*).

There is one critical point at $y = 0$, and it is a sink (stable).

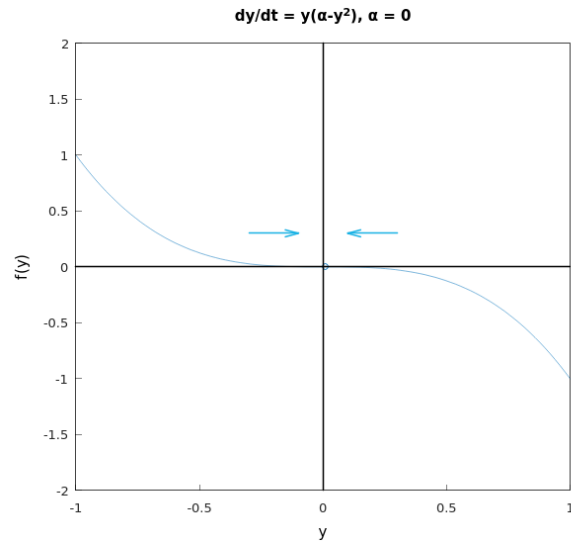


Figure 5.28: The plot for $f(y) = y(\alpha - y^2)$ with $\alpha = 0$, the phase line is drawn with the blue arrows (*Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 26 Only 1 Equilibrium Solution/main.cpp*).

There is one critical point at $y = 0$, and it is a sink (stable).

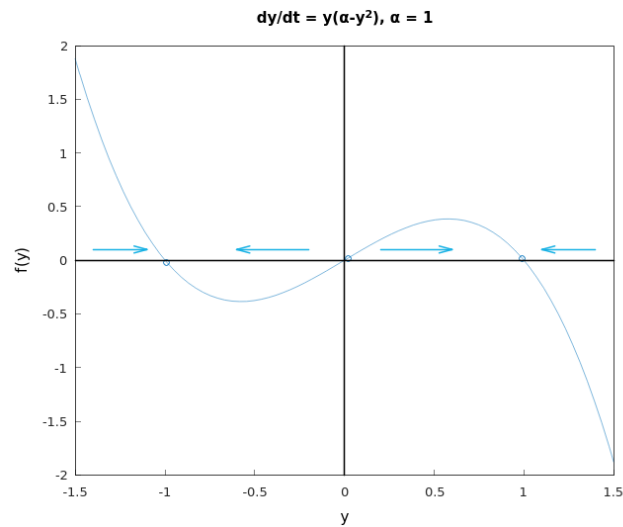


Figure 5.29: The plot for $f(y) = y(\alpha - y^2)$ with $\alpha = 1$, the phase line is drawn with the blue arrows (*Hamzstplot/Examples with Makefile/Plot 2D for Bifurcation Points Problem 26 2 Equilibrium Solution-s/main.cpp*).

There are now three critical points at $y = -\sqrt{a}$, $y = 0$, and $y = \sqrt{a}$, and they are stable, unstable, and stable, respectively.

(b) We will draw the solutions in the ty -plane / the direction field.

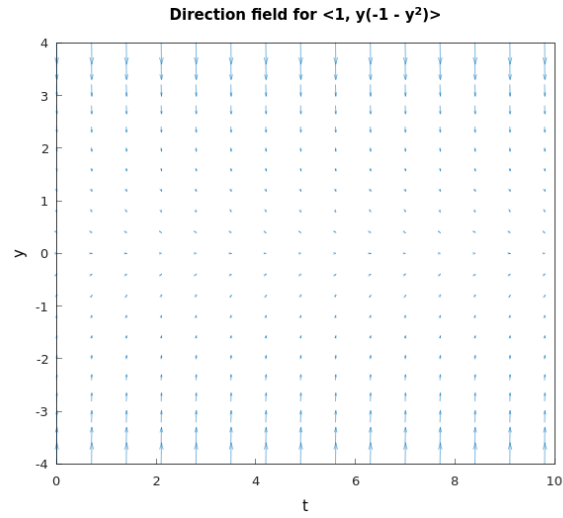


Figure 5.30: The direction field for $\frac{dy}{dt} = y(-1 - y^2)$. Solution curves in the ty -plane corresponding to $\alpha = 1$ are tangent to the vectors in the direction field $\langle 1, y(-1 - y^2) \rangle$ at every point (Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp).

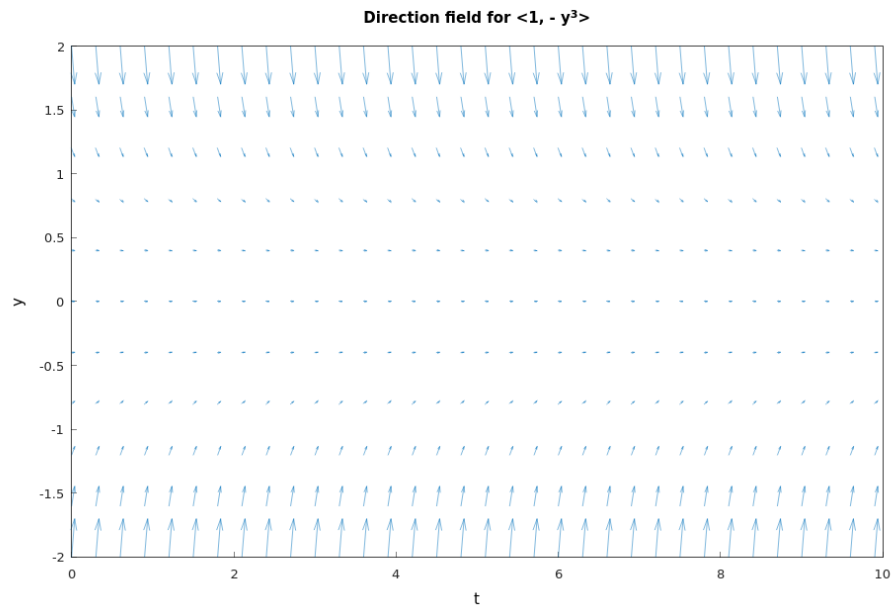


Figure 5.31: The direction field for $\frac{dy}{dt} = -y^3$. Solution curves in the ty -plane corresponding to $\alpha = 0$ are tangent to the vectors in the direction field $\langle 1, -y^3 \rangle$ at every point (Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp).

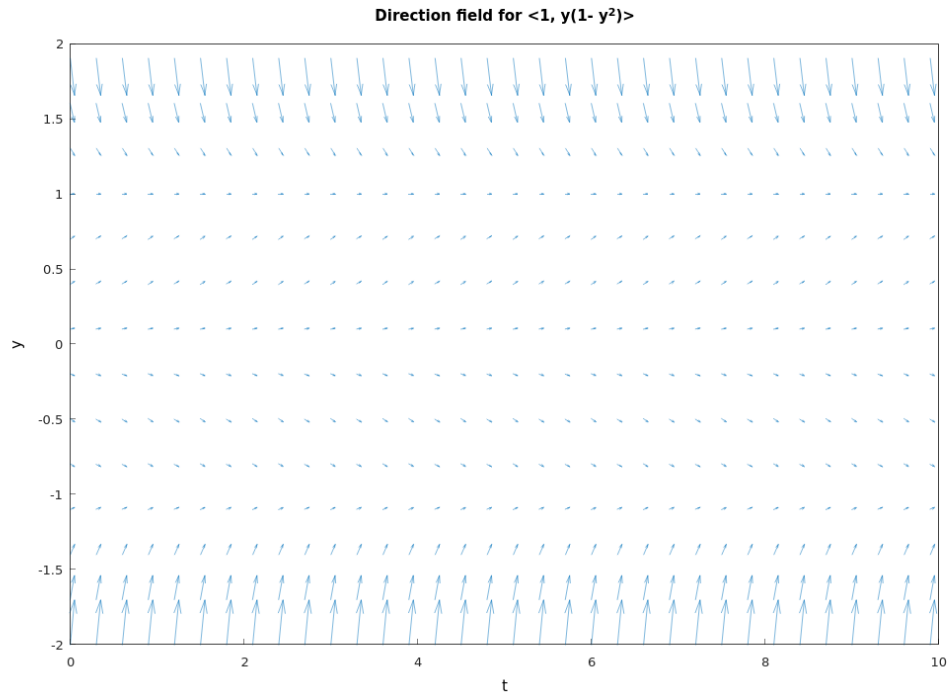


Figure 5.32: The direction field for $\frac{dy}{dt} = y(1 - y^2)$. Solution curves in the ty -plane corresponding to $\alpha = 1$ are tangent to the vectors in the direction field $\langle 1, y(1 - y^2) \rangle$ at every point (Hamzstplot/Examples with Makefile/Plot Direction Fields for Bifurcation Points Problem 25c/main.cpp).

(c) This is the bifurcation diagram, looks like a pitchfork.

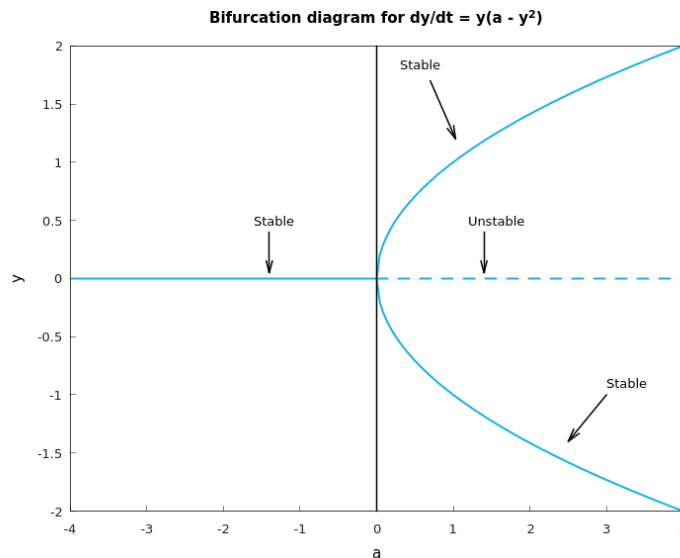


Figure 5.33: The bifurcation diagram for $\frac{dy}{dt} = y(a - y^2)$, the bifurcation point at $a = 0$ is called a saddle-node bifurcation where the solution then differ into two kind of solution: stable and unstable (Hamzstplot/Examples with Makefile/Plot 2D Bifurcation Diagram Problem 26/main.cpp).

[H*] Example:

Consider the equation

$$\frac{dy}{dt} = ay - y^2 = y(a - y) \quad (5.73)$$

- Consider the cases $a < 0$, $a = 0$, and $a > 0$. In each case find all of the critical points for Eq. (4.73), draw the phase line, and determine whether each critical point is asymptotically stable, semistable, or unstable.
- In each case sketch several solutions of Eq. (4.73) in the ty -plane.
- Draw the bifurcation diagram of Eq. (4.73). Observe that for Eq. (4.73) there are the same number of critical points for $a < 0$ and $a > 0$ but that their stability has changed. For $a < 0$ the equilibrium solution $y = 0$ is asymptotically stable and $y = a$ is unstable, while for $a > 0$ the situation is reversed. Thus there has been an exchange of stability as a passes through the bifurcation point $a = 0$. This type of bifurcation is called a transcritical bifurcation.

Solution:

- .

[H*] Chemical Reactions Example:

A second order chemical reaction involves the interaction (collision) of one molecule of a substance P with one molecule of a substance Q to produce one molecule of a new substance X ; this is denoted by



Suppose that p and q , where $p \neq q$, are the initial concentrations of P and Q , respectively, and let $x(t)$ be the concentration of X at time t . Then $p - x(t)$ and $q - x(t)$ are the concentrations of P and Q at time t , and the rate at which the reaction occurs is given by the equation

$$\frac{dx}{dt} = \alpha(p - x)(q - x) \quad (5.74)$$

where α is a positive constant.

- If $x(0) = 0$, determine the limiting value of $x(t)$ as $t \rightarrow \infty$ without solving the differential equation. Then solve the initial value problem and find $x(t)$ for any t .
- If the substances P and Q are the same, then $p = q$ and Eq. (4.74) is replaced by

$$\frac{dx}{dt} = \alpha(p - x)^2 \quad (5.75)$$

If $x(0) = 0$, determine the limiting value of $x(t)$ as $t \rightarrow \infty$ without solving the differential equation. Then solve the initial value problem and determine $x(t)$ for any t .

Solution:

- The chemical reaction for this problem is



The (stoichiometric) coefficients of P and Q are both 1, so the molecules of P and Q combine in a one-to-one fashion. A sample direction field

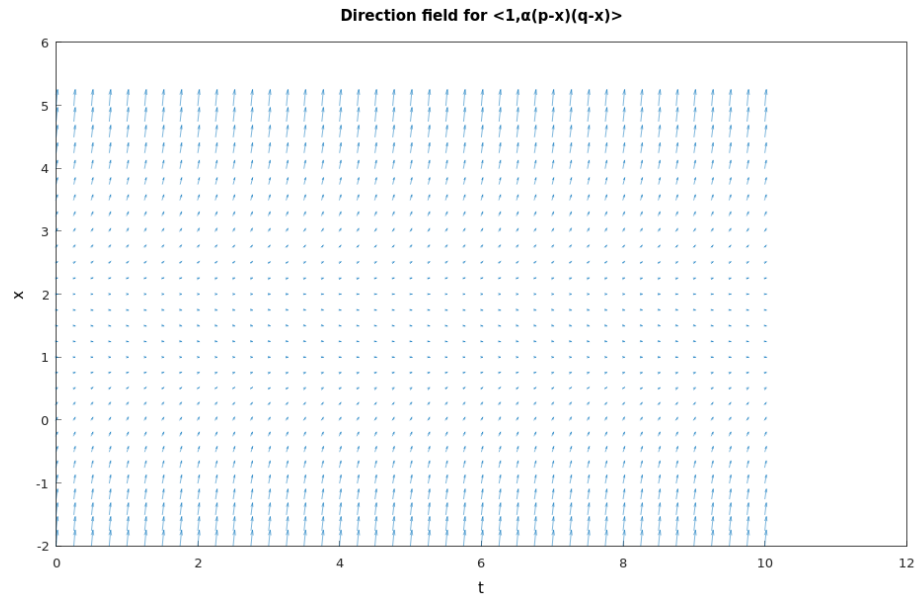


Figure 5.34: The direction field for $\langle 1, \alpha(p-x)(q-x) \rangle$ with $\alpha = 1$, $p = 1$, and $q = 2$ (Hamzstplot/Examples with Makefile/Plot Direction Fields for Chemical Reactions Problem 28a/main.cpp).

We see that the solution $x(t)$ tends to 1 as $t \rightarrow \infty$ for the initial condition $x(0) = 0$. We conclude then that $x(t)$ tends to p or q , whichever of the two is smaller, as $t \rightarrow \infty$.

$$\lim_{t \rightarrow \infty} x(t) = \begin{cases} p, & \text{if } p < q \\ q, & \text{if } q < p \end{cases}$$

Solve the ODE now by separating variables

$$\begin{aligned}
 \frac{dx}{dt} &= \alpha(p-x)(q-x) \\
 \frac{dx}{(p-x)(q-x)} &= \alpha dt \\
 \int \frac{dx}{(p-x)(q-x)} &= \alpha t + C \\
 \int \left(\frac{1/(q-p)}{p-x} + \frac{1/(p-q)}{q-x} \right) dx &= \alpha t + C \\
 \frac{1}{q-p} \int \frac{dx}{p-x} + \frac{1}{p-q} \int \frac{dx}{q-x} &= \alpha t + C \\
 \frac{1}{p-q} \int \frac{dx}{x-p} + \frac{1}{q-p} \int \frac{dx}{x-q} &= \alpha t + C \\
 \frac{1}{p-q} \ln|x-p| + \frac{1}{q-p} \ln|x-q| &= \alpha t + C \\
 \ln|x-p|^{\frac{1}{p-q}} + \ln|x-q|^{\frac{1}{q-p}} &= \alpha t + C \\
 -\ln|x-p|^{\frac{1}{q-p}} + \ln|x-q|^{\frac{1}{q-p}} &= \alpha t + C \\
 \ln \left| \frac{x-q}{x-p} \right|^{\frac{1}{q-p}} &= \alpha t + C
 \end{aligned}$$

Apply the initial condition $x(0) = 0$ now to determine C

$$\begin{aligned}
 \ln \left| \frac{-q}{-p} \right|^{\frac{1}{q-p}} &= C \\
 C &= \ln \left| \frac{q}{p} \right|^{\frac{1}{q-p}}
 \end{aligned}$$

The previous equation is then

$$\begin{aligned}
 \ln \left| \frac{x-q}{x-p} \right|^{\frac{1}{q-p}} &= \alpha t + \ln \left| \frac{q}{p} \right|^{\frac{1}{q-p}} \\
 \frac{1}{q-p} \ln \left| \frac{x-q}{x-p} \right| &= \alpha t + \frac{1}{q-p} \ln \left| \frac{q}{p} \right| \\
 \ln \left| \frac{x-q}{x-p} \right| &= \alpha t(q-p) + \ln \left| \frac{q}{p} \right| \\
 \left| \frac{x-q}{x-p} \right| &= e^{\alpha t(q-p)} \left(\frac{q}{p} \right) \\
 \frac{x-q}{x-p} &= \pm \frac{q}{p} e^{\alpha t(q-p)} \\
 x-q &= \frac{q}{p} e^{\alpha t(q-p)} x - q e^{\alpha t(q-p)} \\
 x \left[1 - \frac{q}{p} e^{\alpha t(q-p)} \right] &= q - q e^{\alpha t(q-p)} \\
 x(t) &= \frac{q(1 - e^{\alpha t(q-p)})}{1 - \frac{q}{p} e^{\alpha t(q-p)}}
 \end{aligned}$$

from the $\pm$ sign, we choose the plus sign so that the initial condition remains satisfied. Thus, we simplify again one time and the general solution can be written as

$$x(t) = \frac{pq(1 - e^{\alpha t(q-p)})}{p - qe^{\alpha t(q-p)}} \quad (5.76)$$

- (b) The direction field below shows that the solution curves, which lie tangent to the direction field vectors at every point, show that the solution $x(t)$ tends to 1 as $t \rightarrow \infty$ with the initial value of $x(0) = 0$ and $p = 1, \alpha = 1$. We conclude that

$$\lim_{t \rightarrow \infty} x(t) = p$$

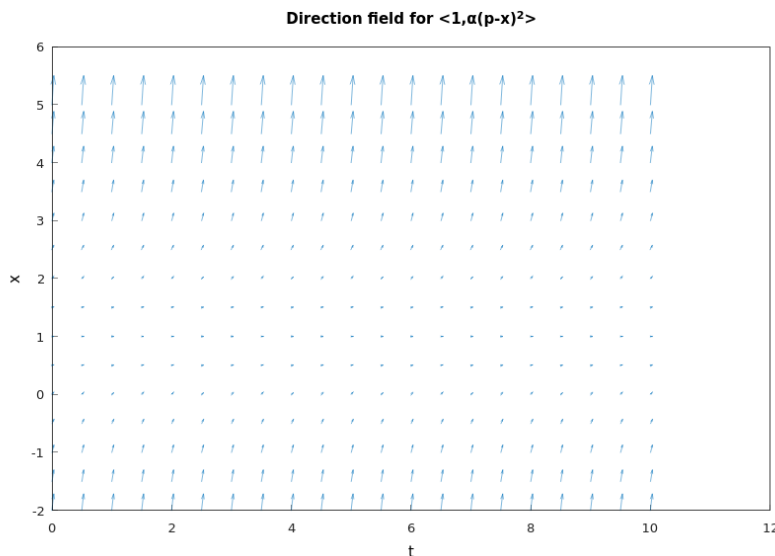


Figure 5.35: The direction field for $\langle 1, \alpha(p-x)^2 \rangle$ with $\alpha = 1$ and $p = 1$ (Hamzstplot/Examples with Makefile/Plot Direction Fields for Chemical Reactions Problem 28b/main.cpp).

Solve the ODE now by separating variables

$$\begin{aligned} \frac{dx}{dt} &= \alpha(p-x)^2 \\ \frac{dx}{(p-x)^2} &= \alpha dt \\ \int \frac{dx}{(p-x)^2} &= \alpha t + C_1 \\ \int \frac{dx}{(x-p)^2} &= \alpha t + C_1 \\ -\frac{1}{x-p} &= \alpha t + C_1 \end{aligned}$$

Apply the initial condition $x(0) = 0$ now to determine C_1 .

$$-\frac{1}{-p} = C_1$$

$$C_1 = \frac{1}{p}$$

As a result, the previous equation becomes

$$-\frac{1}{x-p} = \alpha t + \frac{1}{p}$$

$$-(x-p) = \frac{1}{\alpha t + \frac{1}{p}}$$

$$x-p = -\frac{1}{\alpha t + \frac{1}{p}}$$

$$x(t) = p - \frac{1}{\alpha t + \frac{1}{p}}$$

$$= p - \frac{p}{\alpha p t + 1}$$

$$= \frac{\alpha p^2 t + p - p}{\alpha p t + 1}$$

$$x(t) = \frac{\alpha p^2 t}{\alpha p t + 1}$$

- Exact Equations and Integrating Factors

[H*] Exact equations are a class of equations for which there is also a well-defined method of solution.

[H*] **Example:**

Solve the differential equation

$$2x + y^2 + 2xyy' = 0 \quad (5.77)$$

The equation is neither linear nor separable, so the methods suitable for those types of equations are not applicable here. However, observe that the function $\psi(x, y) = x^2 + xy^2$ has the property that

$$2x + y^2 = \frac{\partial \psi}{\partial x}$$

$$2xy = \frac{\partial \psi}{\partial y} \quad (5.78)$$

Therefore the differential equation can be written as

$$\frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y} \frac{dy}{dx} = 0 \quad (5.79)$$

Assuming that y is a function of x and calling upon the chain rule, we can write Eq. (4.79) in the equivalent form

$$\frac{\partial \psi}{\partial x} = \frac{d}{dx}(x^2 + xy^2) = 0 \quad (5.80)$$

Therefore

$$\psi(x, y) = x^2 + xy^2 = c \quad (5.81)$$

where c is an arbitrary constant, is an equation that defines solutions of Eq. (4.77) implicitly.

The key step here was the recognition that there is a function ψ that satisfies Eqs. (4.78).

[H*] More generally, let the differential equation

$$M(x, y) + N(x, y)y' = 0 \quad (5.82)$$

be given. Suppose that we can identify a function ψ such that

$$\begin{aligned} \frac{\partial \psi}{\partial x}(x, y) &= M(x, y) \\ \frac{\partial \psi}{\partial y}(x, y) &= N(x, y) \end{aligned} \quad (5.83)$$

and such that $\psi(x, y) = c$ defines $y = \phi(x)$ implicitly as a differentiable function of x . Then

$$M(x, y) + N(x, y)y' = \frac{\partial \psi}{\partial x} + \frac{\partial \psi}{\partial y} \frac{dy}{dx} = \frac{d}{dx} \psi[x, \phi(x)]$$

and the differential equation (4.82) becomes

$$\frac{d}{dx} \psi[x, \phi(x)] = 0 \quad (5.84)$$

In this case Eq. (4.82) is said to be an exact differential equation. Solutions of Eq. (4.82), or the equivalent Eq. (4.84), are given implicitly by

$$\psi(x, y) = c \quad (5.85)$$

where c is an arbitrary constant.

Theorem 5.3: Exact Differential Equation in a Region R

Let the functions M, N, M_y , and N_x , where subscripts denote the partial derivatives, be continuous in the rectangular region $R : \alpha < x < \beta, \gamma < y < \delta$. Then Eq. (4.82)

$$M(x, y) + N(x, y)y' = 0$$

is an exact differential equation in R if and only if

$$M_y(x, y) = N_x(x, y) \quad (5.86)$$

at each point of R . That is, there exists a function ψ satisfying Eqs. (4.83),

$$\psi_x(x, y) = M(x, y)$$

$$\psi_y(x, y) = N(x, y)$$

if and only if M and N satisfy Eq. (4.86).

- Numerical Approximations: Euler's Method

[H*] Numerical methods for ordinary differential equations are methods used to find numerical approximations to the solutions of ordinary differential equations. Their use is also known as "numerical integration."

Many differential equations cannot be solved exactly. For practical purposes, however - such as in engineering - a numeric approximation to the solution is often sufficient. The algorithms studied here can be used to compute such an approximation. An alternative method is to use techniques from calculus to obtain a series expansion of the solution.

Ordinary differential equations occur in many scientific disciplines, including physics, chemistry, biology, and economics. In addition, some methods in numerical partial differential equations convert the partial differential equation into an ordinary differential equation, which must then be solved.

[H*] **Euler Method**

From any point on a curve, you can find an approximation of a nearby point on the curve by moving a short distance along a line tangent to the curve.

Suppose we have a first-order differential equation with initial value problem of the form

$$y'(t) = \frac{dy}{dt} = f(t, y(t)), \quad y(t_0) = y_0 \quad (5.87)$$

Starting with the differential equation Eq. (4.76), we replace the derivative y' by the finite difference approximation

$$y'(t) \approx \frac{y(t+h) - y(t)}{h} \quad (5.88)$$

which when re-arranged yields the following formula

$$y(t+h) \approx y(t) + hy'(t)$$

and using Eq. (4.76) gives

$$y(t+h) \approx y(t) + hf(t, y(t)) \quad (5.89)$$

This formula is usually applied in the following way.

We choose a step size h , and we construct the sequence t_0, t_1

$$\begin{aligned} t_0 \\ t_1 &= t_0 + h \\ t_2 &= t_0 + 2h \\ &\vdots \end{aligned}$$

Then we denote y_n , as the numerical estimate of the exact solution $y(t_n)$.

We compute these estimates by the following recursive scheme

$$y_{n+1} = y_n + hf(t_n, y_n) \quad (5.90)$$

This is the Euler method (or forward Euler method). The method is named after Leonhard Euler who described it in 1768.

The Euler method is an example of an explicit method. This means that the new value y_{n+1} is defined in terms of things that are already known, like y_n .

[H*] Backward Euler Method

If, instead of Eq. (4.71), we use the approximation

$$y'(t) \approx \frac{y(t) - y(t-h)}{h} \quad (5.91)$$

we get the backward Euler method:

$$y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}) \quad (5.92)$$

The backward Euler method is an implicit method, meaning that we have to solve an equation to find y_{n+1} . One often uses fixed-point iteration or the Newton-Raphson method to achieve this.

It costs more time to solve this equation than explicit methods; this cost must be taken into consideration when one selects the method to use.

The advantage of implicit methods are that they usually more stable for solving a stiff equation, meanin that a larger step size h can be used.

The limitation to the Euler method is that it can be prone to significant errors, especially with larger step sizes or highly nonlinear functions. Furthermore, the Euler method is often not accurate enough. In more precise terms, it only has order one. The first-order accuracy means that the error is proportional to the step size h . This caused mathematicians to look for higher-order methods.

One possibility is to use not only the previously computed value y_n to determine y_{n+1} , but to make the solution depend on more past values. This yields a so-called multistep method. Another possibility is to use more points in the interval $[t_n, t_{n+1}]$. This leads to the family of Runge-Kutta methods. One of their fourth-order methods is especially popular.

[H*] Numerical analysis is not only the design of numerical methods, but also their analysis. Three central concepts in this analysis are:

1. Convergence: whether the method approximates the solution.

A numerical method is said to be convergent if the numerical solution approaches the exact solution as the step size h goes to 0. More precisely, we require that for every ordinary differential equations with type of Eq. (4.76) with a Lipschitz function f and every $t^* > 0$,

$$\lim_{h \rightarrow 0^+} \max_{n=0,1,\dots,[t^*/h]} ||y_{n,h} - y(t_n)|| = 0$$

2. Order: how well it approximates the solution.

Suppose the numerical method is

$$y_{n+k} = \psi(t_{n+k}; y_n, y_{n+1}, \dots, y_{n+k-1}; h)$$

The local (truncation) error of the method is the error committed by one step of the method. That is, it is the difference between the result given by the method, assuming that no error was made in earlier steps, and the exact solution:

$$\delta_{n+k}^h = \psi(t_{n+k}; y_n, y_{n+1}, \dots, y_{n+k-1}; h) - y(t_{n+k})$$

The method is said to be consistent if

$$\lim_{h \rightarrow 0} \frac{\delta_{n+k}^h}{h} = 0$$

The method has order p if

$$\delta_{n+k}^h = O(h^{p+1})$$

as $h \rightarrow 0$.

Hence a method is consistent if it has an order greater than 0. The forward Euler method and the backward Euler method both have order 1, so they are consistent. Most methods being used in practice attain higher order. Consistency is a necessary condition for convergence, but not sufficient; for a method to be convergent, it must be both consistent and zero-stable.

A related concept is the global(truncation) error, the error sustained in all the steps one needs to reach a fixed time t . Explicitly, the global error at time t is $y_N - y(t)$ where $N = \frac{t-t_0}{h}$. The global error of a p th order one-step method is $O(h^p)$; in particular, such a method is convergent. This statement is not necessarily true for multi-step methods.

3. Stability: whether errors are damped out.

For some differential equations, application of standard methods-such as the Euler method, explicit Runge-Kutta methods, multistep methods (e.g., Adams-Bashforth methods)-exhibit instability in the solutions, though other methods may produce stable solutions.

This "difficult behavior" in the equation (which may not necessarily be complex itself) is described as stiffness, and is often caused by the presence of different time scales in the underlying problem.

For example, a collision in a mechanical system like in an impact oscillator typically occurs at much smaller time scale than the time for the motion of objects; this discrepancy makes for very "sharp turns" in the curves of the state parameters.

Stiff problems are ubiquitous in chemical kinetics, control theory, solid mechanics, weather forecasting, biology, plasma physics, and electronics. One way to overcome stiffness is to extend the notion of differential equation to that of differential inclusion, which allows for and models non-smoothness.

V. SECOND ORDER LINEAR EQUATIONS

- Homogeneous Equations with Constant Coefficients

[H*] The first
[H*]

- Solutions of Linear Homogeneous Equations; the Wronskian

[H*] The first
[H*]

VI. HIGHER ORDER LINEAR EQUATIONS

- General Theory of n th Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

VII. SERIES SOLUTIONS OF SECOND ORDER LINEAR EQUATIONS

- Power Series

[H*] The first
[H*]

- Series Solutions Near an Ordinary Point

[H*] The first
[H*]

VIII. LAPLACE TRANSFORM

- Definition

[H*] The first
[H*]

- Solution of Initial Value Problems

[H*] The first
[H*]

IX. SYSTEMS OF FIRST ORDER LINEAR EQUATIONS

- [Introduction](#)

[\[H*\]](#) The first
[\[H*\]](#)

- [Linear Algebraic Equations](#)

[\[H*\]](#) The first
[\[H*\]](#)

X. NUMERICAL METHODS

- The Runge-Kutta Method

[H*] The first
[H*]

- Multistep Methods

[H*] The first
[H*]

XI. NONLINEAR DIFFERENTIAL EQUATIONS AND STABILITY

- General Theory of n th Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

XII. BOUNDARY VALUE PROBLEMS AND STURM-LIOUVILLE THEORY

- The Occurrence of Two-Point Boundary Value Problems

[H*] The first
[H*]

- Nonhomogeneous Boundary Value Problems

[H*] The first
[H*]

Chapter 6

Partial Differential Equations

"PDE makes my Mathematics Learning more colorful..." - Glanz

"Learn from basic Ordinary Differential Equation, don't jump to PDE yet (but still help Glanz learning PDE on 2023)" - Sentinel

There is always a funny story behind every chapter in Lastthrim Projection, it is only 3 days before Partial Differential Equation exam, and after surgery on Glanz' left leg, miss 2 weeks of classes, she thinks a chapter dedicated for PDE(Partial Differential Equation) is better than include just a snippet in the **Elementary Differential Equation** chapter, actually Freya is the one ordered her to add this chapter. We shall see why this chapter exists, the book bought by Glanz has her drawing of Stacy (Valhalla Projection' subordinate in Coppito, L'Aquila and Todai / University of Tokyo).

There are a lot of books talking about Partial Differential Equations, a lot of people wrote in their blogs and websites, and create the video to explain it. We read and watch a lot then try to summarize it after we able to comprehend. Hopefully, after certain time if we read it again, the memory will be refreshed quickly.

I. PRELIMINARIES

[H*] In many important physical problems there are two or more independent variables governing the process, thus the corresponding mathematical models involve partial differential equations. For solving partial differential equations we use a method separation of variables, with its essential feature to replace the partial differential equation by a set of ordinary differential equations, which then will be solved by the given initial or boundary conditions.

The desired solution of the partial differential equation is expressed as a sum, or an infinite series, formed from solutions of the ordinary differential equations.

We will call Partial Differential Equation with PDE, interchangeably.

[H*]

II. DERIVATION OF PARTIAL DIFFERENTIAL EQUATIONS

[H*] Partial differential equations arise in many physical problems. Three known partial differential equations are Heat equation, wave equation, and Laplace's equation.

[H*] The key defining property of a partial differential equation (PDE) [?] is that

- There is more than one independent variable $x, y, \dots$
- There is a dependent variable that is an unknown function of these variables $u(x, y, \dots)$
- $\partial u / \partial x$ denoted as the partial derivative of the dependent variable toward the independent variable x .

[H*] When a solution function has two or more independent variables, say $u(x, y)$, then this function can make a partial differential equation by specifying a relation between its' derivatives [?], such as

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Another way of writing partial differential equation is to see it as an identity that relates the independent variables, the dependent variable u , and the derivatives of u . It can be written as

$$F(x, y, u, u_x, u_y) = 0 \quad (6.1)$$

with $u = u(x, y)$, a function in x and y terms as the independent variables.

The general second-order PDE in two independent variables is

$$F(x, y, u, u_x, u_y, u_{xx}, u_{xy}, u_{yy}) = 0 \quad (6.2)$$

A solution of a PDE is a function $u(x, y, \dots)$ that satisfies the equation identically.

[H*] For PDE, the distinction between the independent variables and the dependent variable (the unknown / the solution) is always maintained.

[H*] An example of a general linear second-order partial differential equation is like this:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} = 0$$

The general nonlinear second-order partial differential equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} + u^2 = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial t} + \sin(u) = 0$$

The differences are:

1. The linear partial differential equation will contain the unknown / the solution with power of 1, the nonlinear will contain the unknown / the solution with power other than 1 (could be 2 or argument of $\exp, \log, \sin$).

[H*] A linear partial differential equation is homogeneous if the unknown or the solution that satisfies the partial differential equation is 0, for example:

$$\frac{\partial^2 u}{\partial t^2} + c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

and the solution of the partial differential equation above is $u(x, t) = 0$, thus the partial differential equation is homogeneous.

A homogeneous linear differential equation has an important property:

$$\alpha u + \beta v \quad (6.3)$$

are solutions for $\alpha, \beta \in \mathcal{R}$, whenever u and v are solutions of the equation.

A linear partial differential equation is non-homogeneous if the unknown or the solution that satisfies the partial differential equation is not 0, for example:

$$\frac{\partial u}{\partial t} = 2u + \sin(4\pi t)$$

is equivalent to

$$\frac{\partial u}{\partial t} - 2u = \sin(4\pi t)$$

When all terms involving the unknown / the solution on the left side of equation and the right side is not 0 then it is non-homogeneous

[H*] If u and v is any function and $\mathcal{L}$ is an operator, then $\mathcal{L}v$ is a new function. Thus the definition for linearity is

$$\mathcal{L}(u + v) = \mathcal{L}u + \mathcal{L}v \quad \mathcal{L}(cu) = c(\mathcal{L}u) \quad (6.4)$$

Hence

$$\mathcal{L}(u) = 0 \quad (6.5)$$

is called a homogeneous linear equation. Then

$$\mathcal{L}(u) = g \quad (6.6)$$

where $g \neq 0$ is a given function of the independent variables, is called a non-homogeneous linear equation, for any functions u, v and any constant c .

[H*] Recall this Taylor series expansion of a function $f(z)$ about the point $z = a$, the expansion is given by

$$f(z) = f(a) + f'(a)(z - a) + \frac{f''(a)}{2!}(z - a)^2 + \frac{f'''(a)}{3!}(z - a)^3 + \dots$$

where $f'(a)$ means the derivative of $f(z)$ with respect to z evaluated at the point $z = a$, $f''(a)$ is the second derivative of $f(z)$ with respect to z evaluated at the point $z = a$, etc. If, in the above equation we let

$$z = a + h \quad \text{and} \quad a = x$$

we obtain

$$f(x + h) = f(x) + f'(x)h + \dots$$

Thus the value of $f(x)$ at the point $x + h$ (near to x) is given by the value of the function at the point x plus the value of its first derivative evaluated at x , multiplied by the distance $[(x + h) - x = h]$ between the two points.

[H*] The partial derivative of a function $u(x, y)$ with respect to x at the point x , is denoted by

$$\frac{\partial u}{\partial x}$$

[H*] The general solution of the partial differential equation involves as many arbitrary functions as the order of the equation (the order of the highest partial differential coefficient in the equation), for example:

$$\frac{\partial u(x, t)}{\partial t} = -D \frac{\partial^2 u(x, t)}{\partial x^2}$$

has order of 2, we can also call it second-order partial differential equation, this form of PDE is known as the Heat Equation.

[H*] If $w(x, y, z, t)$ is a function of time and the three rectangular coordinates of a point in space, then the following are partial differential equations with order of 2:

$$\begin{aligned} \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} &= 0 \\ a^2 \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) &= \frac{\partial w}{\partial t} \\ a^2 \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) &= \frac{\partial^2 w}{\partial t^2} \end{aligned}$$

The first equation is called Laplace's equation, the second is the heat equation in 3-dimension, and the third is the wave equation in 3-dimension.

[H*] The conditions of partial differential equations are classified into two categories [?]:

1. Initial Value Problem (IVP)

A partial differential equation with initial conditions will have a dependent variable and a sufficient number of its derivatives are prescribed at the initial point of domain.

2. Boundary Value Problem (BVP)

A partial differential equations with its dependent variable and a sufficient number of its derivatives are prescribed at the boundary of domain.

For example consider this heat equation or a metal rod:

$$\frac{\partial u(x, t)}{\partial t} = -D \frac{\partial^2 u(x, t)}{\partial x^2}, \quad 0 \leq x \leq 20, \quad t > 0$$

with $u(x, t)$ denotes the temperature of the rod at length- x at time t . The length of the rod is 20 cm.

A lot of people sometimes write u to replace $u(x, t)$, it could cause confusion to those newly learning differential equation, we prefer to write the whole as $u(x, t)$ to denote all the dependent variables that is used by the function u . Just like when we first learning calculus we write $f(x)$ not only writing f .

Now for the initial condition, it will be written like this:

$$u(x, 0) = \begin{cases} x, & 0 \leq x \leq 10 \\ 20 - x, & 10 \leq x \leq 20 \end{cases}$$

and the boundary conditions:

$$\begin{aligned} u(0, t) &= 0 \\ u(L, t) &= 0 \end{aligned}$$

both ends of the metal rod will be cooled at 0^0 perhaps attaching both ends at ice cube. From this we can learn how the temperature will behave on the rod after certain period of time.

[H*] Consider this PDE:

$$\frac{\partial^2 u(x, t)}{\partial t^2} - \frac{\partial^2 u(x, t)}{\partial x^2} = 0, \quad 0 \leq x \leq L, \quad t > 0$$

There are four broad categories of boundary conditions:

1. Dirichlet boundary conditions

On the boundary, the values of the dependent variable are specified.

$$u(0, t) = 0, \quad u(L, t) = 0$$

2. Neumann boundary conditions

On the boundary, the normal derivative of the dependent variable is specified.

$$\frac{\partial u(0, t)}{\partial x} = 0, \quad \frac{\partial u(L, t)}{\partial x} = 0$$

3. Cauchy boundary conditions

On the boundary, both the values of the dependent variable and its normal derivative are specified.

$$u(0, t) = 0, \quad \frac{\partial u(L, t)}{\partial x} = 0$$

4. Robin boundary conditions

On the boundary, a linear combination of the dependent variable and its normal derivatives are specified.

$$u(0, t) = 1, \quad \frac{\partial u(L, t)}{\partial x} + u(L, t) = 0$$

[H*] A second-order partial differential equation of the form

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu < 0$$

is elliptic, while

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$$

is parabolic, and

$$Au_{xx} + Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu > 0$$

is hyperbolic.

[H*] To have a unique, stable solution, each class of PDEs requires a different class of boundary conditions

1. Dirichlet or Neumann boundary conditions on a closed boundary surrounding the region of interest are the requirements to elliptic equations. Other boundary conditions are either insufficient to determine a unique solution, overly restrictive or lead to instabilities.
2. Cauchy boundary conditions on an open surface are the requirements to elliptic equations. Other boundary conditions are either too restrictive for a solution to exist, or insufficient to determine a unique solution.

3. Parabolic equations require Dirichlet or Neumann boundary conditions on an open surface. Other boundary conditions are too restrictive.

- **Two-Point Boundary Value Problems**

[H*] Consider this differential equation of second order

$$y'' + p(t)y' + q(t)y = g(t) \quad (6.7)$$

with the initial conditions

$$y(t_0) = y_0 \quad y'(t_0) = y'_0 \quad (6.8)$$

[H*] Consider this independent variables: $x, y, z, \dots$
and now an unknown function of these variables can be written as below

$$u(x, y, \dots) \quad (6.9)$$

We denote the derivative with subscript, for example a derivative of u with respect to the independent variable x is

$$\frac{\partial u}{\partial x} = u_x \quad (6.10)$$

for y :

$$\frac{\partial u}{\partial y} = u_y \quad (6.11)$$

[H*] **Partial Differential Equation**

A PDE is an identity that relates the independent variables $(x, y, \dots)$ with the dependent variable u , and the partial derivatives of u . It can be written as

$$F(x, y, u(x, y), u_x(x, y), u_y(x, y)) = F(x, y, u, u_x, u_y) = 0 \quad (6.12)$$

This is the most general PDE in two independent variables x and y , of first order.

[H*] **Order**

The order of an equation is the highest derivative that appears.

For example:

- * $u_t + 2u_x$ is a first order partial differential equation
- * $u_{xx} - u_t$ is a second order partial differential equation

[H*] **Solution**

A solution to a PDE is a function $u(x, y, \dots)$ that satisfies the equation identically.

In PDEs, the distinction between the independent variables and the dependent variable is always maintained.

III. FOURIER SERIES

- **General Theory of n th Order Linear Equations**

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

IV. EVEN AND ODD FUNCTIONS

- General Theory of n th Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

V. THE WAVE EQUATION

- General Theory of n th Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

VI. THE HEAT EQUATION

- General Theory of n th Order Linear Equations

[H*] This heat equation is one of the most famous partial differential equation that we will try to simulate, the applications are tremendous, from Black-Scholes equation derived from this, create a long lasting ice cream, to be able to create better computer components will need the comprehension of this equation as well.

[H*]

- The Method of Variation of Parameters

[H*] The first

[H*]

VII. THE LAPLACE EQUATION

- General Theory of n th Order Linear Equations

[H*] The first
[H*]

- The Method of Variation of Parameters

[H*] The first
[H*]

Chapter 7

Probability and Statistics

"LAM VAM RAM YAM HAM OM AH." - 7 Chakras Chanting

D

I. STATISTICS AND DATA ANALYSIS

Statistical methods are used to analyze data from an industrial process (manufacturing, energy sources, drug testing). Statistical methods are involved with gathering of information or **scientific data**, which then will be processed to extract the knowledge from the data and improve the **quality** of the process.

II. VARIABILITY

Life would be simple for scientists if the observed product were always the same and were always on target, there would be no need for statistical methods. Statisticians make use of fundamental laws of probability and statistical inference to draw conclusions about scientific systems.

Information is gathered in the form of **samples**, or collections of **observations**. Samples are collected from **populations**, which are collections of all individuals or individual items of a particular type.

Research scientists gain much from scientific data. Data provide understanding of scientific phenomena.

Chapter 8

Linear and Nonlinear Programming

"My only place is by your side. If you want to carry out your wish, I will follow you. That is my Nature." - Sarah (Suikoden III)

Linear programming is an optimization problem, a problem that is well rooted as a principle underlying the analysis of many complex decision or allocation problems. For example, if we have limited resources and we want to achieve a certain goal, we need to learn to optimize our resources with either linear programming or nonlinear programming depends on our problems at hand. Like how we decide to buy groceries depending on our needs / diet (the goal), and the money we have (the constraints).

Optimization, also known as mathematical programming, collection of mathematical principles and methods used for solving quantitative problems in many disciplines, including physics, biology, engineering, economics, and business. The subject grew from a realization that quantitative problems in manifestly different disciplines have important mathematical elements in common. Because of this commonality, many problems can be formulated and solved by using the unified set of ideas and methods that make up the field of optimization.

Optimization problems typically have three fundamental elements. The first is a single numerical quantity, or objective function, that is to be maximized or minimized. The objective may be the expected return on a stock portfolio, a company's production costs or profits, the time of arrival of a vehicle at a specified destination, or the vote share of a political candidate. The second element is a collection of variables, which are quantities whose values can be manipulated in order to optimize the objective. Examples include the quantities of stock to be bought or sold, the amounts of various resources to be allocated to different production activities, the route to be followed by a vehicle through a traffic network, or the policies to be advocated by a candidate. The third element of an optimization problem is a set of constraints, which are restrictions on the values that the variables can take. For instance, a manufacturing process cannot require more resources than are available, nor can it employ less than zero resources. Within this broad framework, optimization problems can have different mathematical properties. Problems in which the variables are continuous quantities (as in the resource allocation example) require a different approach from problems in which the variables are discrete or combinatorial quantities (as in the selection of a vehicle route from among a predefined set of possibilities).

I. LINEAR PROGRAMMING

II. NONLINEAR PROGRAMMING

Chapter 9

Introduction to Theory of Interest and Financial Mathematics

"In my world everything what would be it wouldn't be, and what wouldn't be it would." (Alice in Wonderland 1951) - Alice

Mathematical finance, also known as quantitative finance and financial mathematics, is a field of applied mathematics, concerned with mathematical modeling of financial markets.

In general, there exist two separate branches of finance that require advanced quantitative techniques: derivatives pricing on the one hand, and risk and portfolio management on the other. Mathematical finance overlaps heavily with the fields of computational finance and financial engineering. The latter focuses on applications and modeling, often by help of stochastic asset models, while the former focuses, in addition to analysis, on building tools of implementation for the models. Also related is quantitative investing, which relies on statistical and numerical models (and lately machine learning) as opposed to traditional fundamental analysis when managing portfolios.

French mathematician Louis Bachelier's doctoral thesis, defended in 1900, is considered the first scholarly work on mathematical finance. But mathematical finance emerged as a discipline in the 1970s, following the work of Fischer Black, Myron Scholes and Robert Merton on option pricing theory. Mathematical investing originated from the research of mathematician Edward Thorp who used statistical methods to first invent card counting in blackjack and then applied its principles to modern systematic investing.

Chapter 10

Introduction to Stochastic Processes

"One side will make you taller, the other side will make you shorter" (Alice in Wonderland 1951) - The Caterpillar

IN probability theory and related fields, a stochastic or random process is a mathematical object usually defined as a sequence of random variables, where the index of the sequence has the interpretation of time. Stochastic processes are widely used as mathematical models of systems and phenomena that appear to vary in a random manner. Examples include the growth of a bacterial population, an electrical current fluctuating due to thermal noise, or the movement of a gas molecule. Stochastic processes have applications in many disciplines such as biology, chemistry, ecology, neuroscience, physics, image processing, signal processing, control theory, information theory, computer science, and telecommunications. Furthermore, seemingly random changes in financial markets have motivated the extensive use of stochastic processes in finance.

Applications and the study of phenomena have in turn inspired the proposal of new stochastic processes. Examples of such stochastic processes include the Wiener process or Brownian motion process, used by Louis Bachelier to study price changes on the Paris Bourse, and the Poisson process, used by A. K. Erlang to study the number of phone calls occurring in a certain period of time. These two stochastic processes are considered the most important and central in the theory of stochastic processes, and were discovered repeatedly and independently, both before and after Bachelier and Erlang, in different settings and countries.

Chapter 11

Numerical Analysis

Je n'avais pas peur, je suis né pour faire ça - Jeanne d'Arc

D

- I. SOLUTIONS OF EQUATIONS IN ONE VARIABLE
- II. INTERPOLATION AND POLYNOMIAL APPROXIMATION
- III. NUMERICAL DIFFERENTIATION
- IV. NUMERICAL INTEGRATION
- V. BOUNDARY-VALUE PROBLEMS FOR ORDINARY DIFFERENTIAL EQUATIONS
- VI. NUMERICAL SOLUTIONS TO PARTIAL DIFFERENTIAL EQUATIONS

Chapter 12

Complex Numbers

"Tobikata Wo Wasureta Tori No You Ni (A Little bird who forgot how to fly)." - Star Ocean 3 OST - MISIA

I. PRELIMINARIES

[H*] Complex analysis, traditionally known as the theory of functions of a complex variable, is the branch of mathematical analysis that investigates functions of complex numbers. It is helpful in many branches of mathematics, including algebraic geometry, number theory, analytic combinatorics, applied mathematics; as well as in physics, including the branches of hydrodynamics, thermodynamics, and particularly quantum mechanics. By extension, use of complex analysis also has applications in engineering fields such as nuclear, aerospace, mechanical and electrical engineering.

As a differentiable function of a complex variable is equal to its Taylor series (that is, it is analytic), complex analysis is particularly concerned with analytic functions of a complex variable (that is, holomorphic functions).

[H*] Complex number arises when no one on this planet up until 2024 A.D. able to compute this:

$$\sqrt{-1}$$

That number is called imaginary number, the notation is $i = \sqrt{-1}$.

[H*] Let $x, y \in \mathbb{R}$, the complex number $z \in \mathbb{C}$ can be defined as

$$z = x + iy \tag{12.1}$$

Some important properties of complex numbers:

- $\operatorname{Re}(z) = x$ and $\operatorname{Im}(z) = y$ are called the real part of z and the imaginary part of z , respectively.
- $|z| = \sqrt{x^2 + y^2}$ is called the modulus (or absolute value) of z .
- $\bar{z} = x - yi$ is called the complex conjugate of z .
- $z\bar{z} = x^2 + y^2 = |z|^2$
- The angle θ is called an argument of z .
- $\operatorname{Re}(z) = |z| \cos \theta$
- $\operatorname{Im}(z) = |z| \sin \theta$

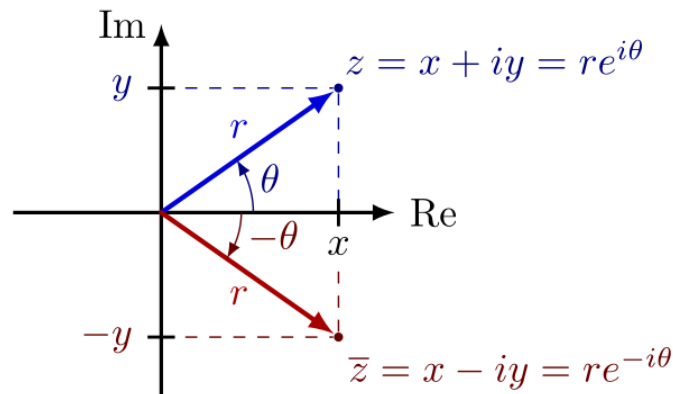


Figure 12.1: The complex number can be represented as $z = x + iy = re^{i\theta}$, with $r = |z|$ and the angle θ is the argument of z .

- $z = |z|(\cos \theta + i \sin \theta)$ is called the polar form of z .

[H*] We try to apply Newton-Raphson method to compute the root of $x^2 + 1 = 0$ and it cannot even compute i .

- [General Theory of \$n\$ th Order Linear Equations](#)

[H*] The first

[H*]

- [The Method of Variation of Parameters](#)

[H*] The first

[H*]

i. C++ Plot: 2D Plot of Complex Function

Suppose we have a complex function

$$e^{ix}$$

and we want to plot it.

Note that

$$e^{ix} = \cos x + i \sin x$$

With **gnuplot** we can plot the function as separate functions, the first function is the real part, the second function is the imaginary part.

Thus, the real part of e^{ix} is $\cos x$ and the imaginary part of e^{ix} is $\sin x$.

```
#include <vector>
#include <cmath>
#include <utility>
#include <boost/tuple/tuple.hpp>
```

```
f(x) = 1+x^2
f'(x) = 2*x
p_{0} = 0.7853982
```

n	p_{n}	p_{n} in Degree
0	0.7853982	45.0
1	-0.24392074	-13.975629
2	1.9278858	110.45972
3	0.7045915	40.37012
4	-0.35733533	-20.473806
5	1.2205782	69.93398
6	0.20064712	11.496233
7	-2.3916135	-137.02936
8	-0.98674273	-56.536194
9	0.013346314	0.7646875
10	-37.456852	-2146.1196
11	-18.715078	-1072.295
12	-9.330823	-534.61676
13	-4.6118255	-264.23813
14	-2.1974957	-125.907234
15	-0.8712162	-49.91701
16	0.13830233	7.9241395
17	-3.5461168	-203.17754
18	-1.6320591	-93.5101
19	-0.509668	-29.201826
20	0.72619677	41.60801
21	-0.32542026	-18.645208
22	1.3737646	78.710915
23	0.3229189	18.50189
24	-1.3869171	-79.4645
25	-0.33294678	-19.076445
26	1.3352681	76.505226
27	0.2931775	16.797832
28	-1.5588628	-89.31626
29	-0.4586848	-26.280704
30	0.8607309	49.316246
31	-0.15053618	-8.625088
32	3.2461925	185.99313
33	1.4690697	84.17149
34	0.3941834	22.585045
35	-1.0713533	-61.384026
36	-0.06897712	-3.9520977
37	7.2142925	413.3485
38	3.5378394	202.70326
39	1.6275904	93.25407
40	0.50659263	29.02562
41	-0.73369	-42.037342

Figure 12.2: The Newton-Raphson method to compute the root of $x^2 + 1 = 0$ cannot converge.

```

#include "gnuplot-iostream.h"

int main() {
    Gnuplot gp;

    // Don't forget to put "\n" at the end of each line!
    gp << "set xrange [-10:5]\nset yrange [-3:3]\n";
    // '-' means read from stdin. The sendld() function sends data to
    // gnuplot's stdin.
    gp << "i = {0,1}\n";
    gp << "plot real(exp(i*x)), imag(exp(i*x))\n";

    // To do parametric plot uncomment below and comment the plot line
    // above
    //gp << "set parametric\n";
    //gp << "plot real(exp(i*t)), imag(exp(i*t))\n";

    return 0;
}

```

Code 18: *main.cpp* "2D plot of complex function"

To compile it, type:

```
g++ -o main main.cpp -lboost_iostreams
./main
```

or with Makefile, type:

```
make
./main
```

s

Explanation for the codes:

- In **gnuplot** complex constants are expressed as $\{ \langle \text{real} \rangle, \langle \text{imag} \rangle \}$, where $\langle \text{real} \rangle$ and $\langle \text{imag} \rangle$ must be numerical constants. For instance:

$$a + bi \rightarrow \{a, b\}$$

$$i \rightarrow \{0, 1\}$$

$$5 - 2i \rightarrow \{5, -2\}$$

Gnuplot can plot only real quantities. Thus we can plot the real part of e^{ix} which is $\cos x$ and the imaginary part of e^{ix} which is $\sin x$, but first we need to define the variable i as $\{0, 1\}$.

```

gp << "i = {0,1}\n";
gp << "plot real(exp(i*x)), imag(exp(i*x))\n";

```

- If you want to do parametric plot with

$$(\text{real}(e^{ix}), \text{imag}(e^{ix}))$$

you can uncomment the two lines at the bottom and comment the plot line above it


```
//gp << "plot real(exp(i*x)), imag(exp(i*x))\n";

// To do parametric plot uncomment below and comment the plot
// line above
gp << "set parametric\n";
gp << "plot real(exp(i*t)), imag(exp(i*t))\n";
```

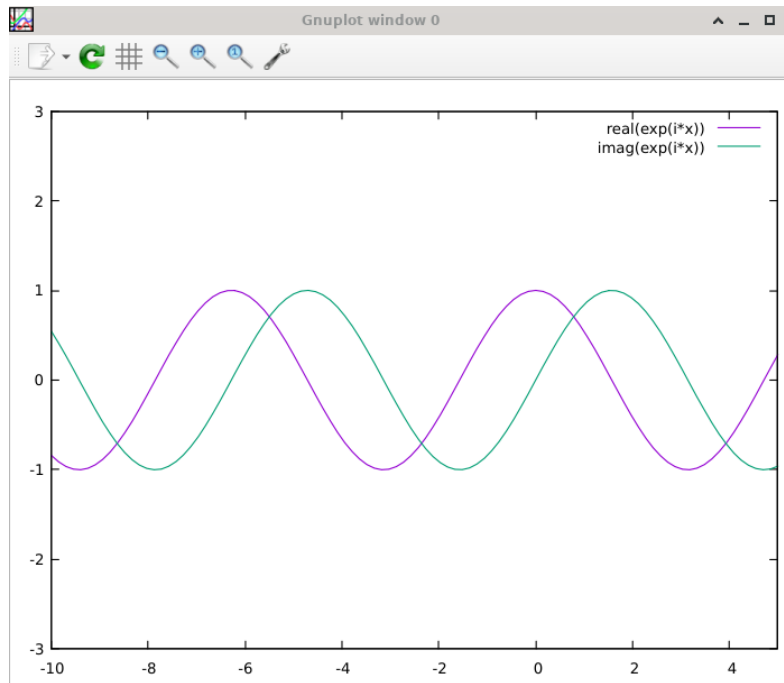


Figure 12.3: The 2D plot of the real part of e^{ix} and the imaginary part of e^{ix} (DFSimulatorC/Source Codes/C++/C++ Gnuplot SymbolicC++/ch23-Numerical Linear Algebra/2D Plot Complex Function/main.cpp).

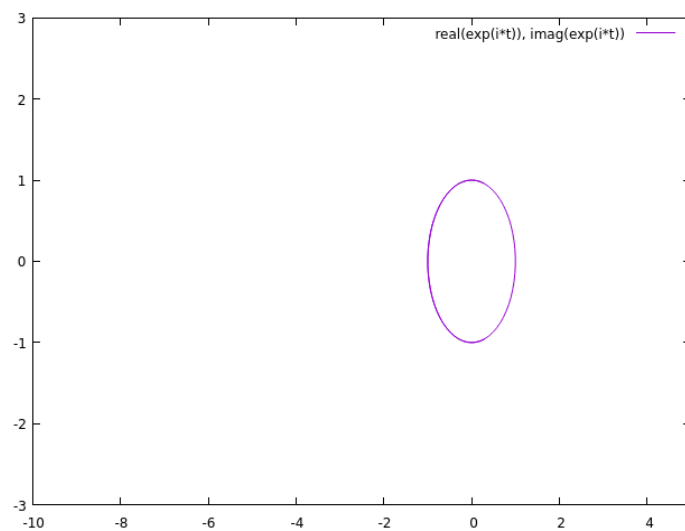


Figure 12.4: The 2D parametric plot of $(\text{real}(e^{ix}), \text{imag}(e^{ix}))$ (DFSimulatorC/Source Codes/C++/C+++ Gnuplot SymbolicC++/ch23-Numerical Linear Algebra/2D Plot Complex Function/main.cpp).

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